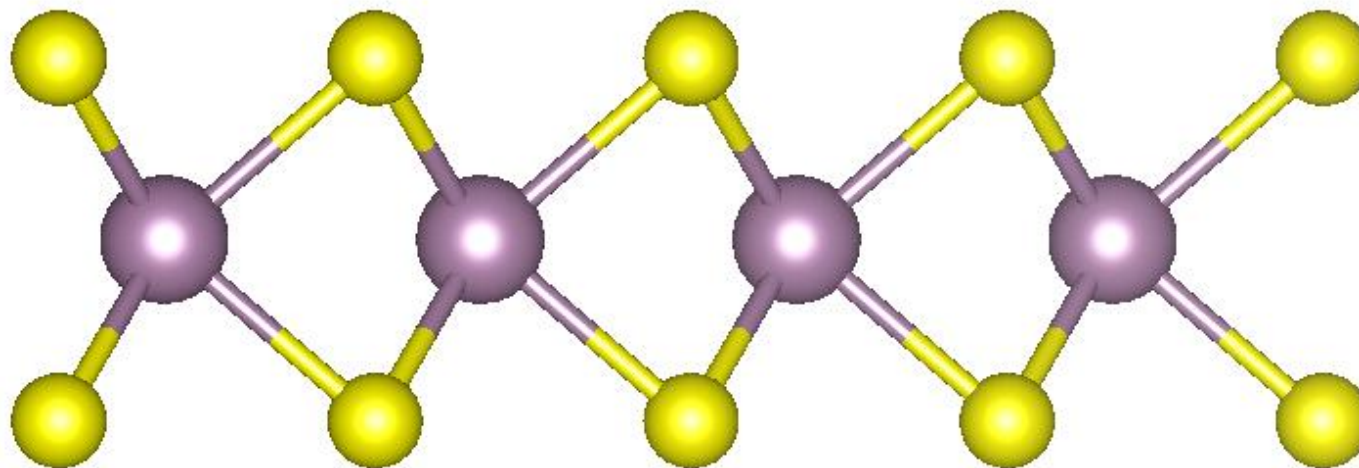




Advancements in Atomic-Scale Control of MoS₂ Etching: Fluorination and Oxidation

Samantha Ellis

Department of Chemistry & Biochemistry, CSUN

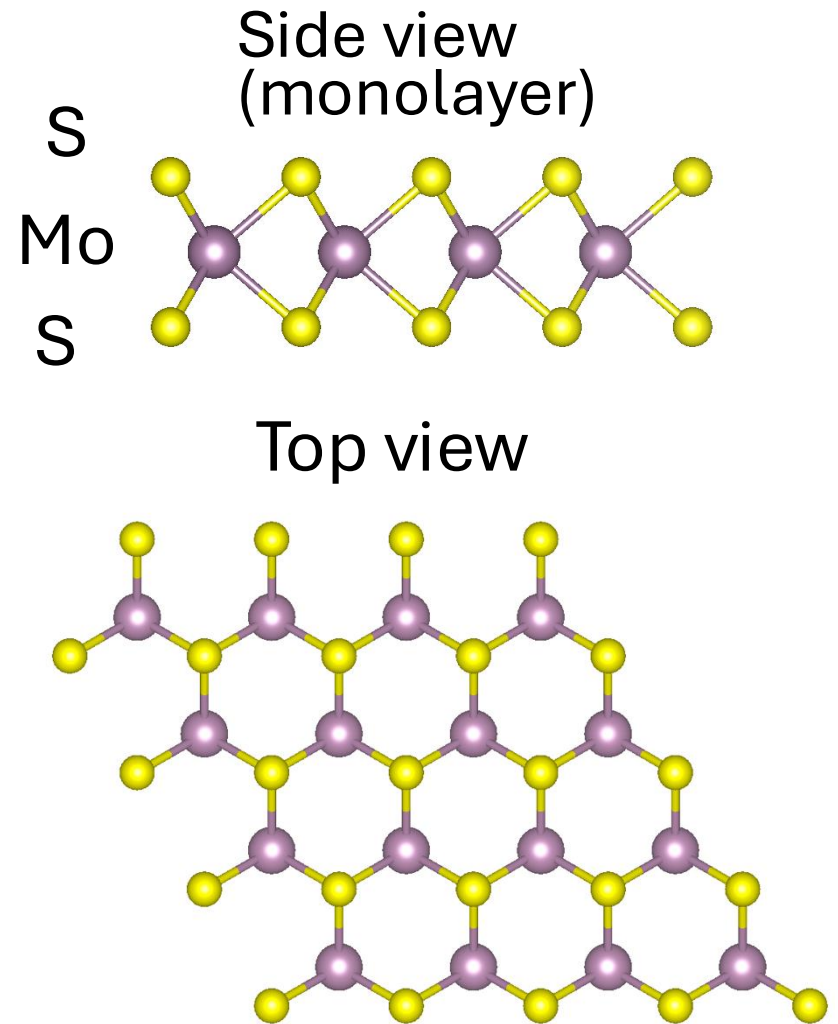


- **Background**
 - Material structure and applications
 - Etching process and control
 - Fluorination vs. oxidation
- Fluorination
 - Atomistic Pathway
 - Controlled Etching
- Oxidation
 - Atomistic Pathway
 - Controlled Etching
- Summary/Outlook

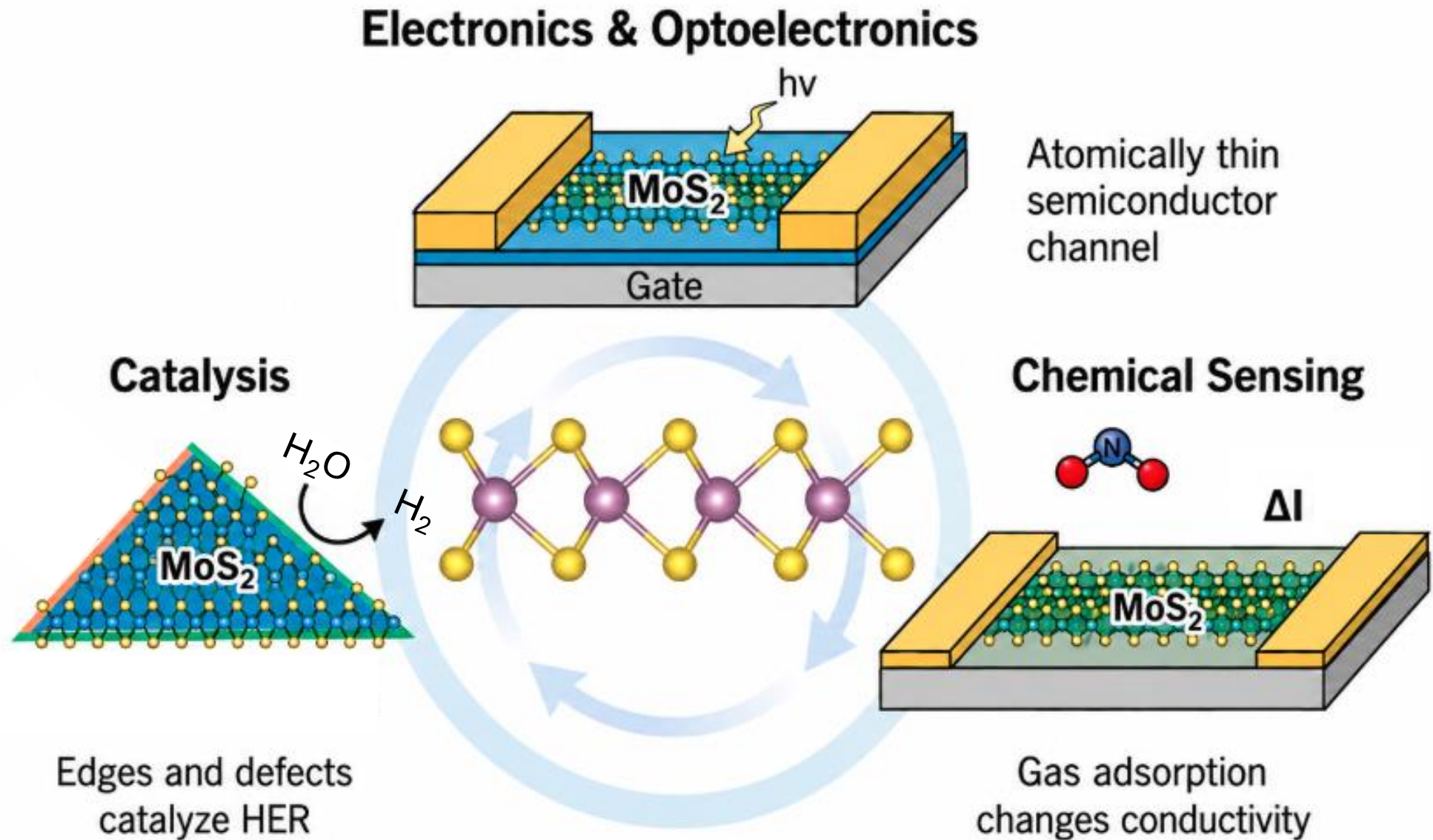
2D transition metal dichalcogenides

- General formula: MX_2
 - M = transition metal (e.g., Mo, W)
 - X = chalcogen (e.g., S, Se, Te)
- Strong covalent bonding in-plane
- Weak van der Waals interactions between layers
- Properties depend on composition, structure, and number of layers

MoS₂: most widely studied
semiconducting 2D TMD



Applications of 2D TMDs



Atomic precision is a necessity

Layer number

Band gap and optical response

Edges and vacancies

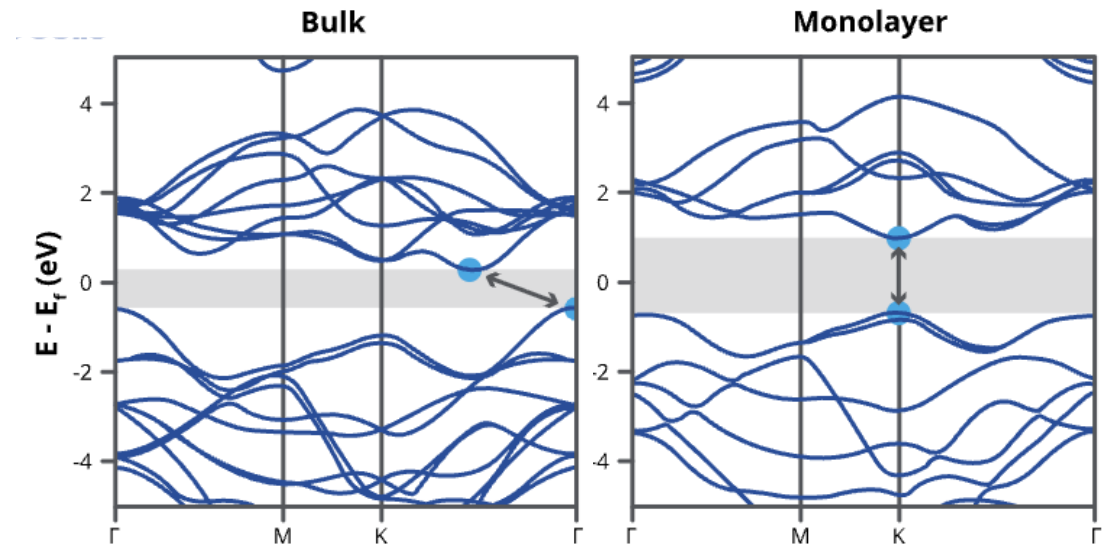
Adsorption strength and catalytic activity

Pattern dimensions and edge roughness

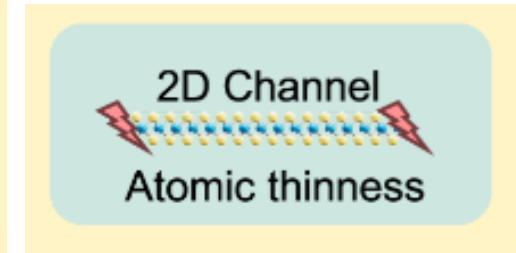
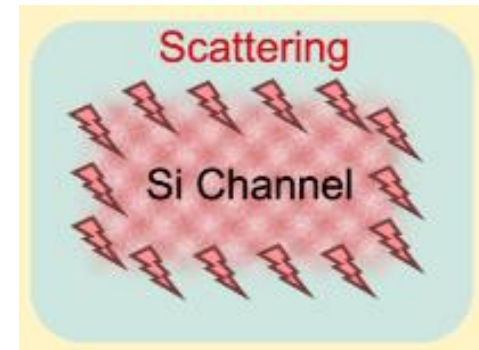
Feature geometry and carrier transport

Residual defects and surface species

Doping, mobility, and stability



Ossila, "Molybdenum Disulfide (MoS_2): Theory & Applications," accessed 2026.

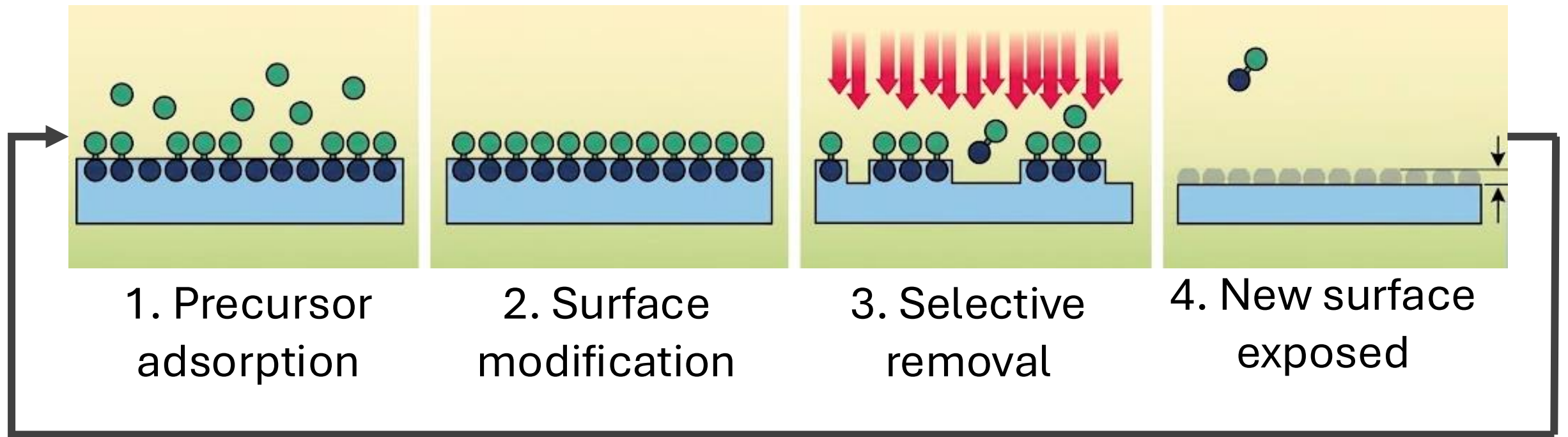


Lan et al., *Nat. Commun.* 17, 7386 (2026), Fig. 1b.

What is etching?

Etching = removal of material from a surface

Ideal Atomic Layer Etching (ALE) sequence



Atomic precision is a necessity

Layer number

Band gap and optical response



Etch exfoliated TMD to yield monolayer

Edges and vacancies

Adsorption strength and catalytic activity



Specific vacancies to promote catalysis

Pattern dimensions and edge roughness

Feature geometry and carrier transport



Allow for microchip design that connects charge carriers

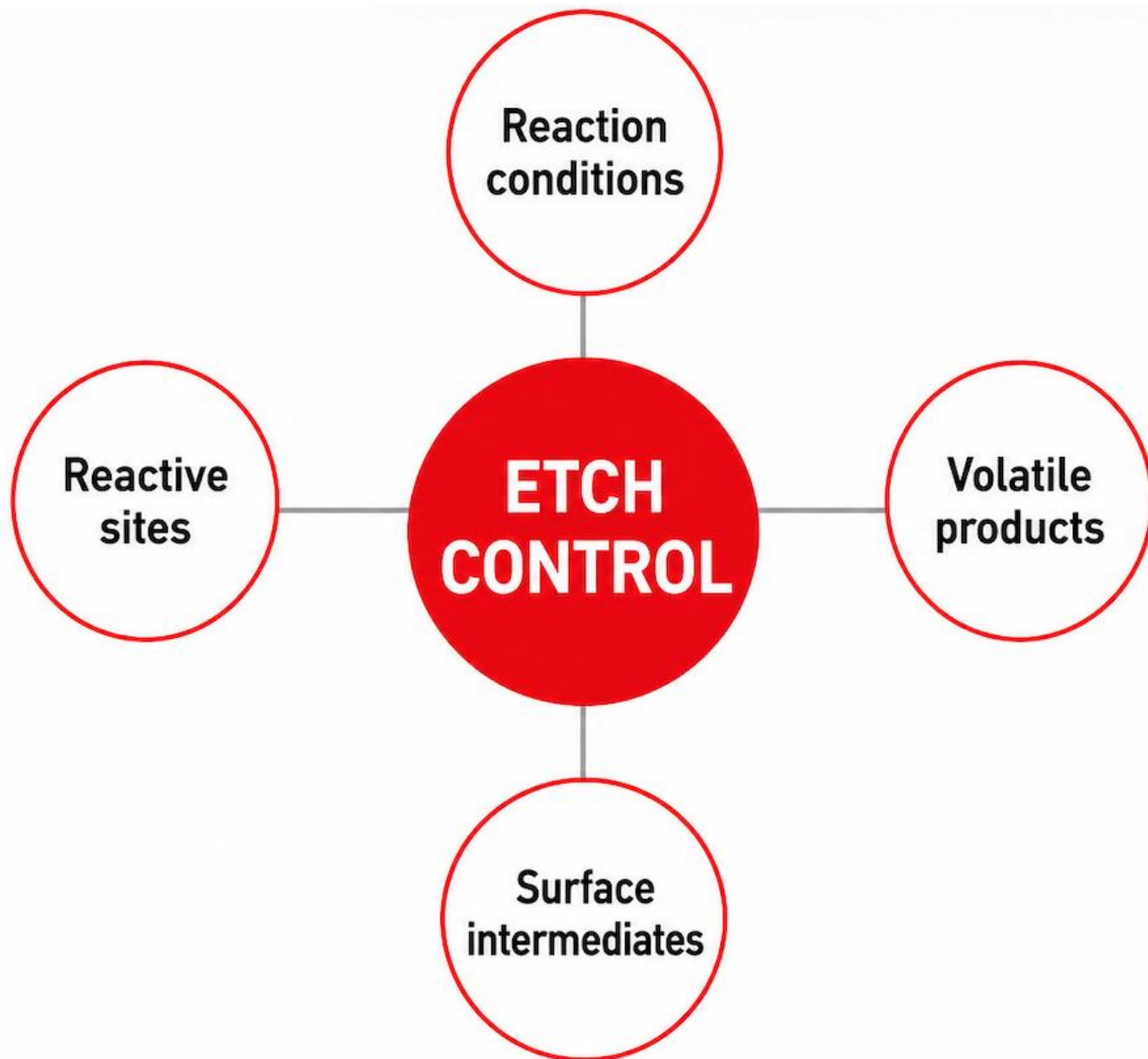
Residual defects and surface species

Doping, mobility, and stability



Prevent chemistry that negatively affects goal properties

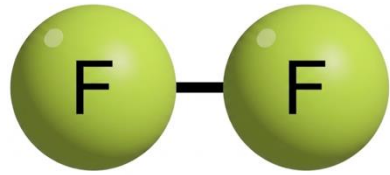
Mechanism guides control



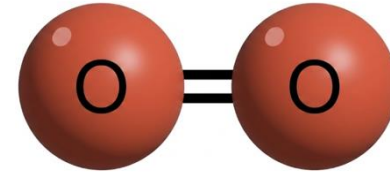
Mechanistic understanding identifies accessible pathways, steps that enable or terminate removal, and conditions that favor controlled etching

Goal: Precise, device-compatible, manufacturable atomic-scale etching

Two routes toward atomic-scale etching



Fluorination vs. Oxidation



Similarities

Gas-phase, chemically driven MoS₂ removal

Reactivity governed by surface state and process conditions

Recent advances span atomistic simulations and cyclic etching experiments

Differences

Surface bonding and modified states

Intermediates, volatile products, and post-reaction surfaces

Removal behavior and control challenges

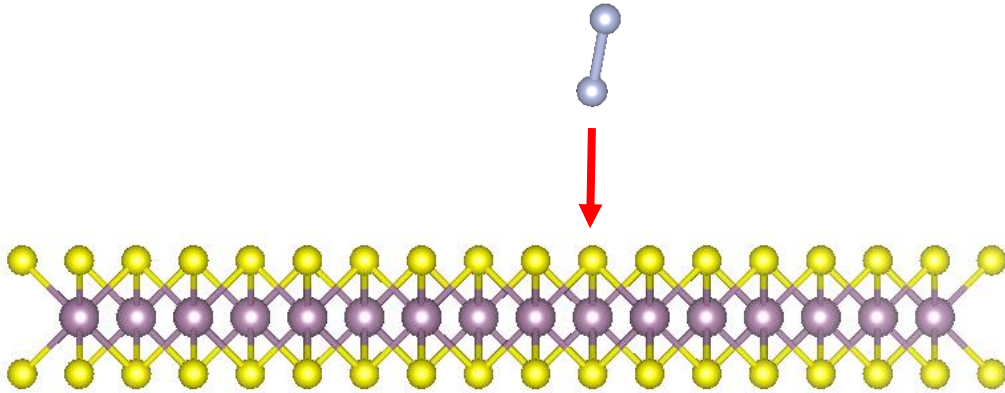
Strategies toward self-limiting ALE

Outline

- Background
 - Material structure and applications
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- **Fluorination**
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 - Controlled Etching
- Oxidation
 - Atomistic Pathway
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- Summary/Outlook

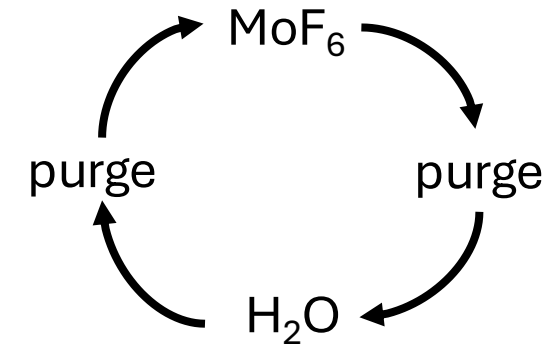
Fluorination mechanisms and etch control

$F_2 + MoS_2 \rightarrow$ atomistic pathway



- How can fluorination remove S and Mo?
- Density Functional Theory (DFT), Ab-Initio Molecular Dynamics (AIMD)
- SF_x formation $\rightarrow V_S \rightarrow MoF_6$ loss

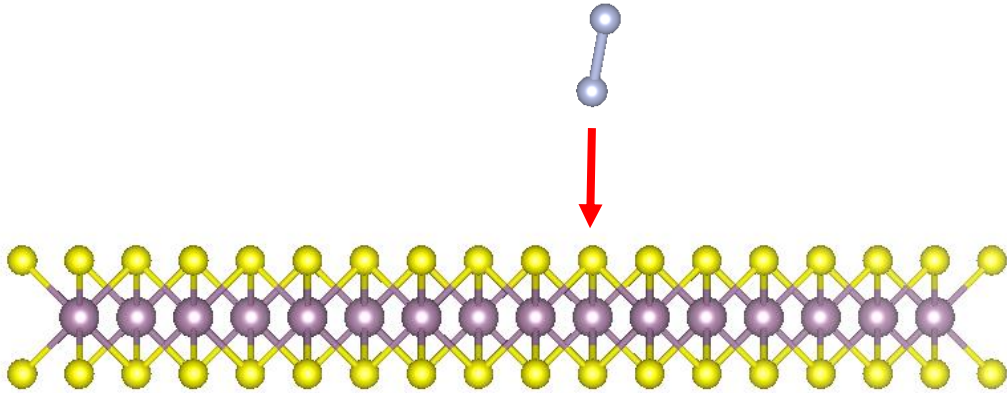
MoF_6/H_2O cycle $\rightarrow$ controlled removal



- Can cyclic exposures produce self-limiting removal?
- Quartz Crystal Microbalance (QCM), Fourier-Transform Infrared Spectroscopy (FTIR), Quadrupole Mass Spectrometry (QMS)
- Saturating half-reactions and alternating O/F surface states

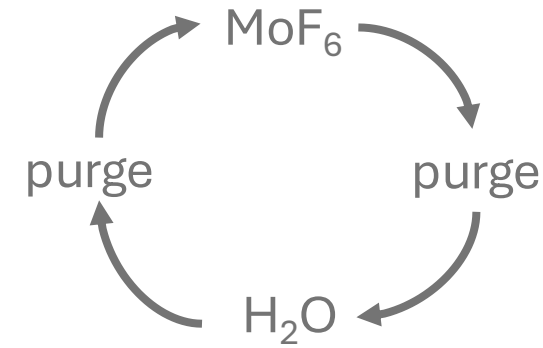
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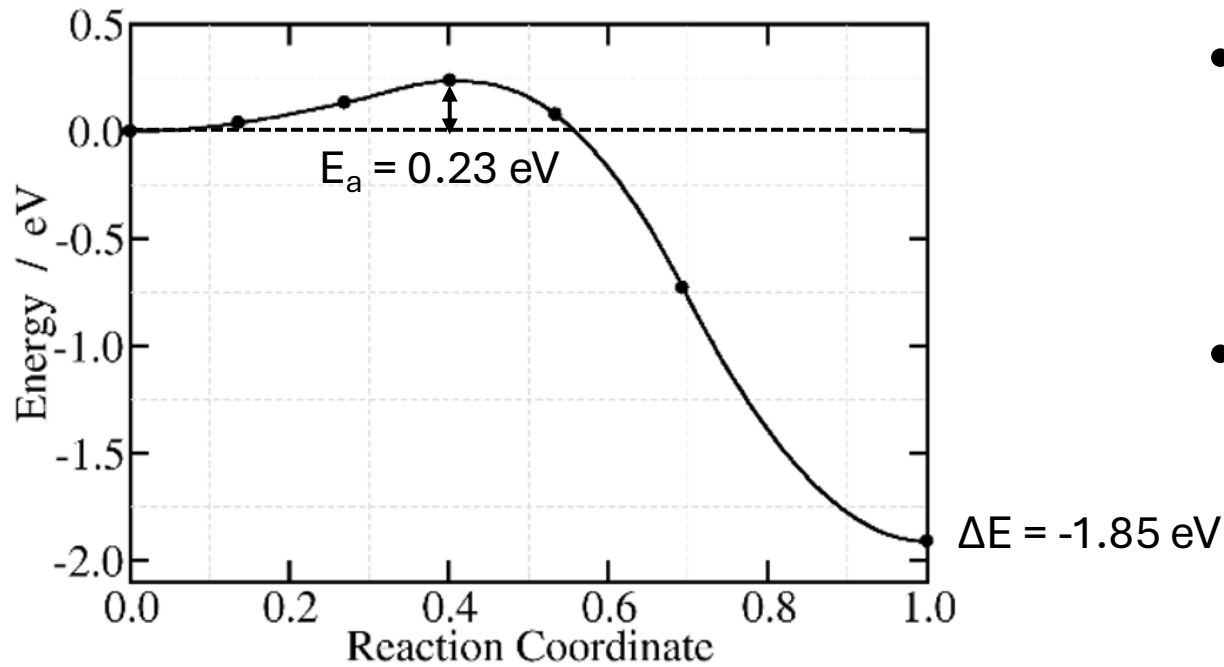
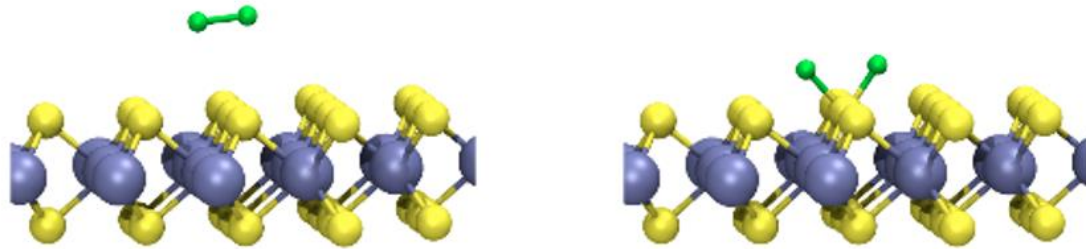
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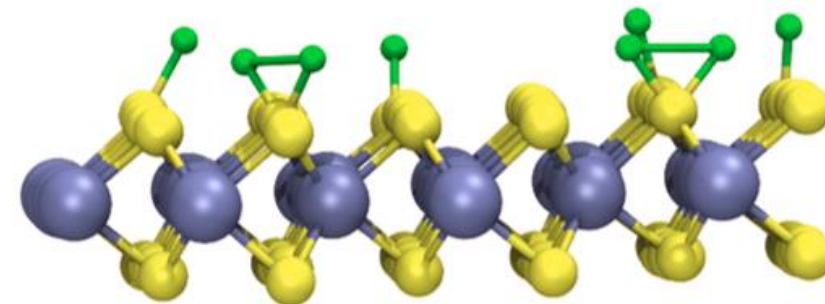


- Can cyclic exposures produce self-limiting removal?
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F_2 readily dissociates on MoS_2

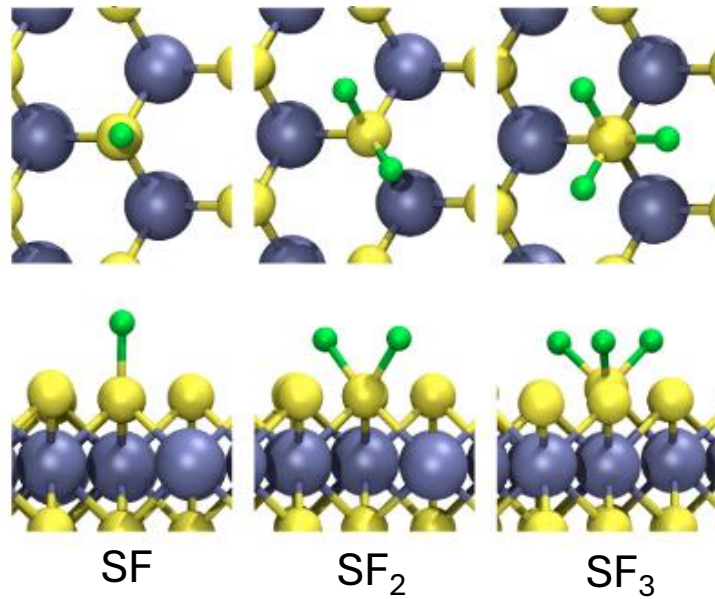


- Adsorbed F_2 dissociates with a low barrier
- Dissociation produces surface SF and SF_2 groups
- Other F sources may enter through different intermediate pathways



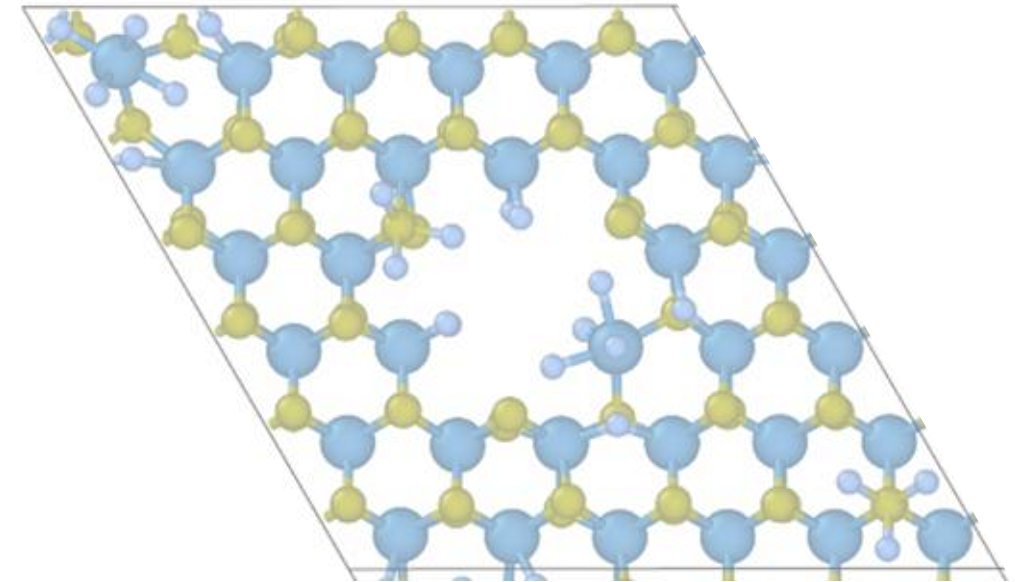
Fluorinated surface after 5 ps

SF₃ desorption depends on site geometry



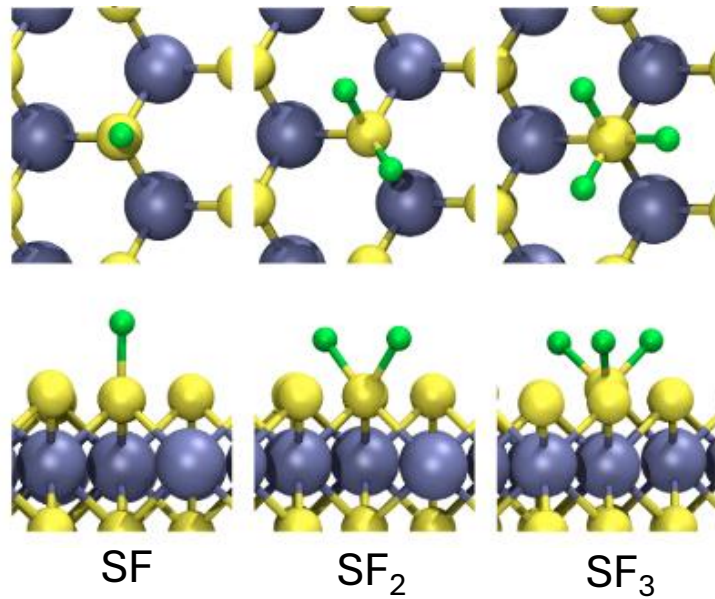
Surface species	ΔE (eV)
S	5.94
SF	3.78
SF ₂	2.08
SF ₃	0.39

Table. Calculated desorption energies of surface SF_x species.



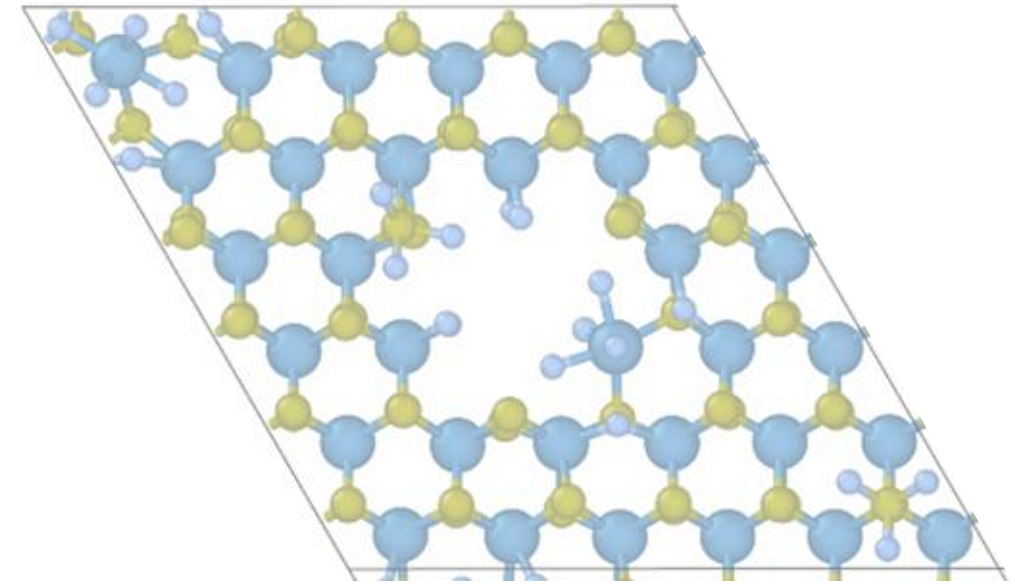
- Edge-bound SF₃ desorbed at **300 K**
- Basal-plane SF₃ persisted through **700 K**
- Pore formation may promote etch propagation

SF₃ desorption depends on site geometry



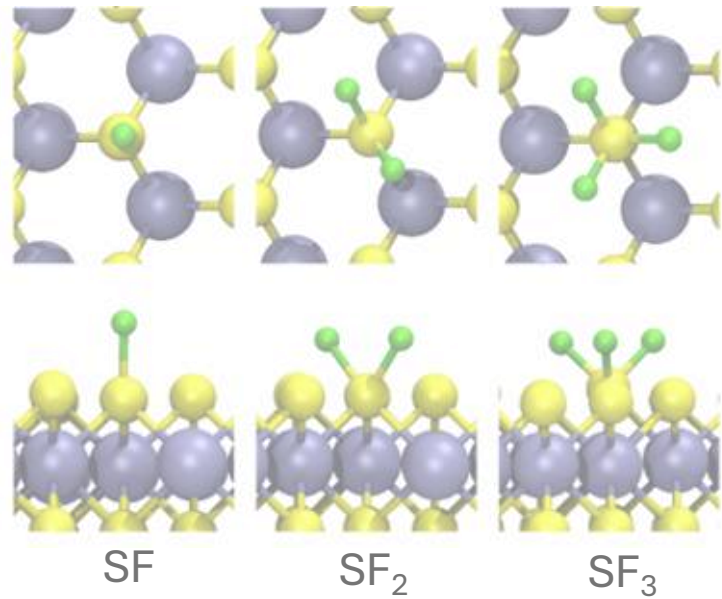
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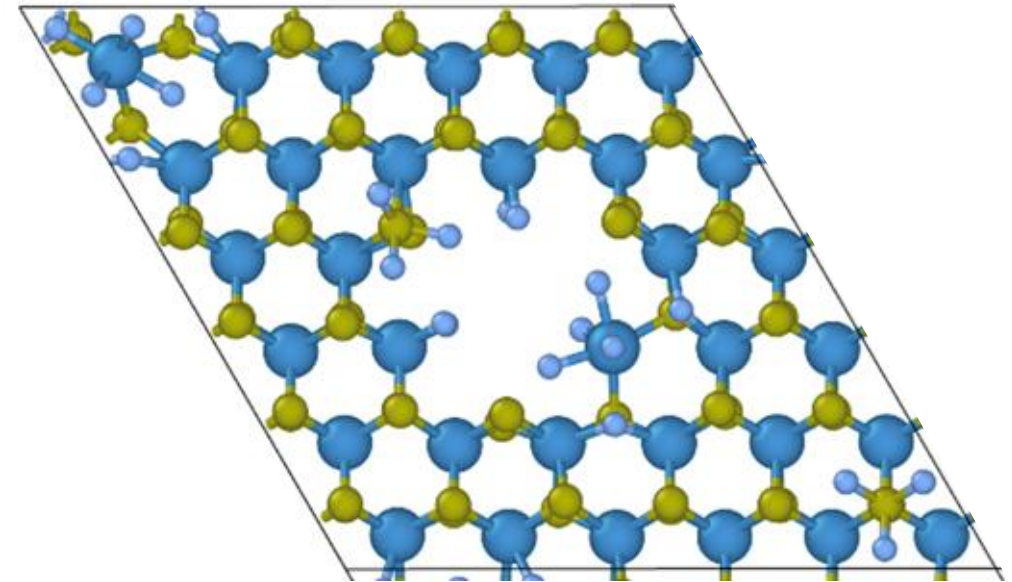
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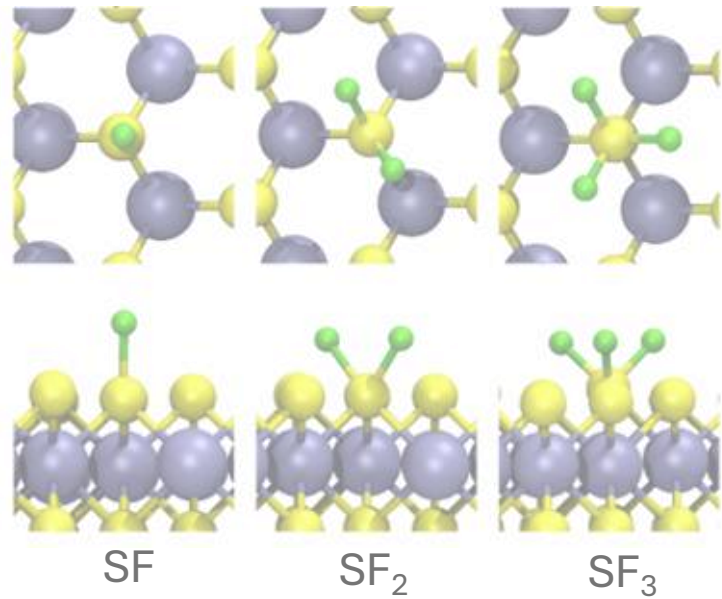
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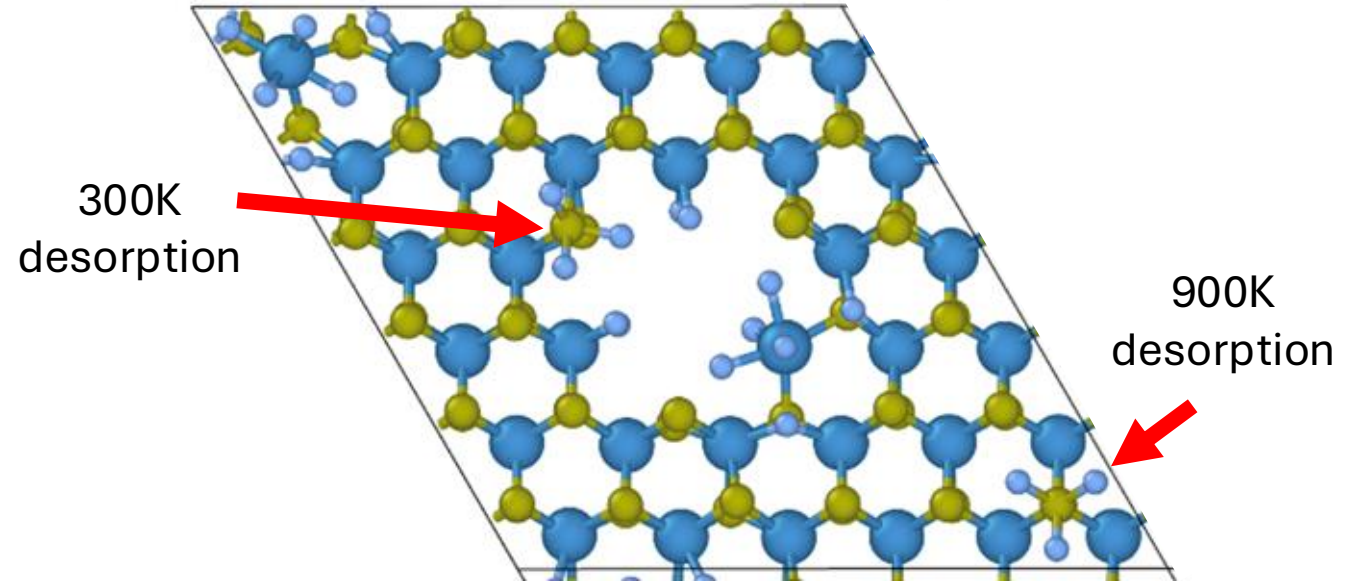
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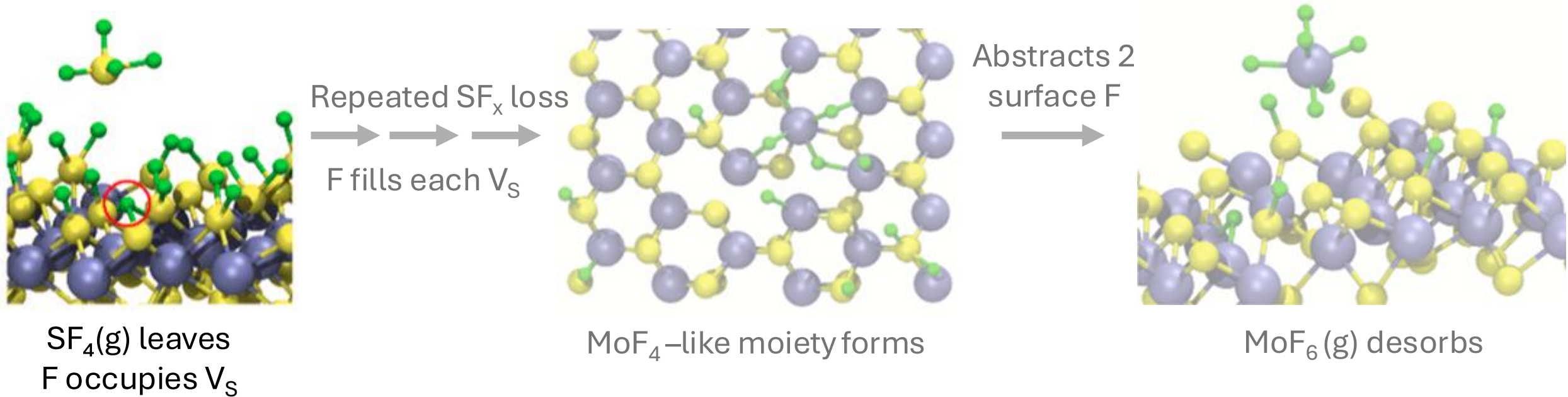
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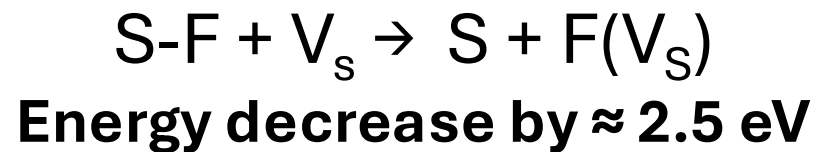
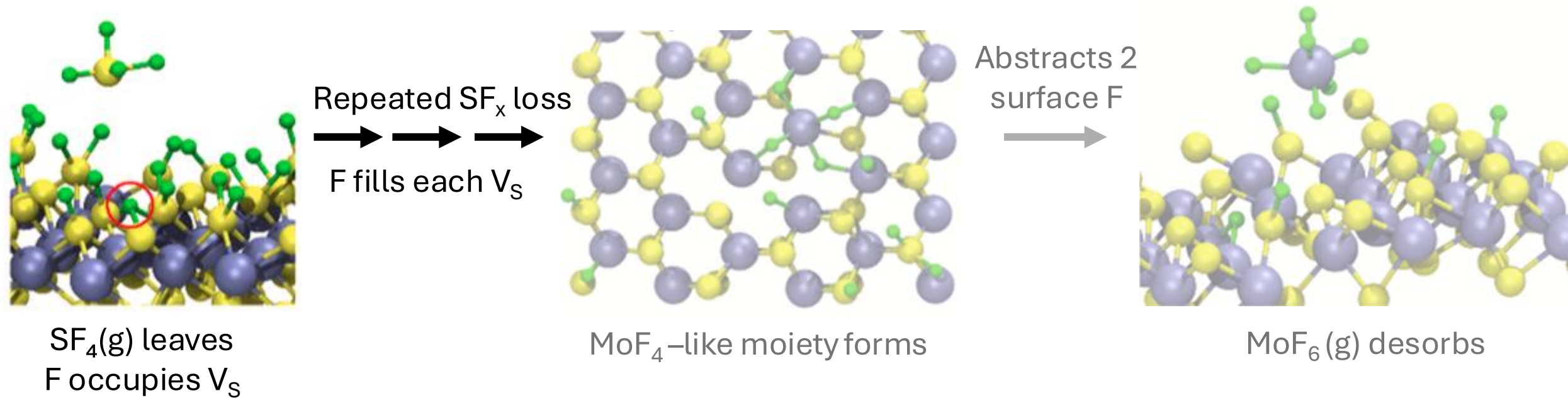


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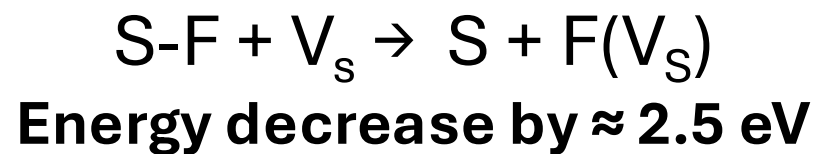
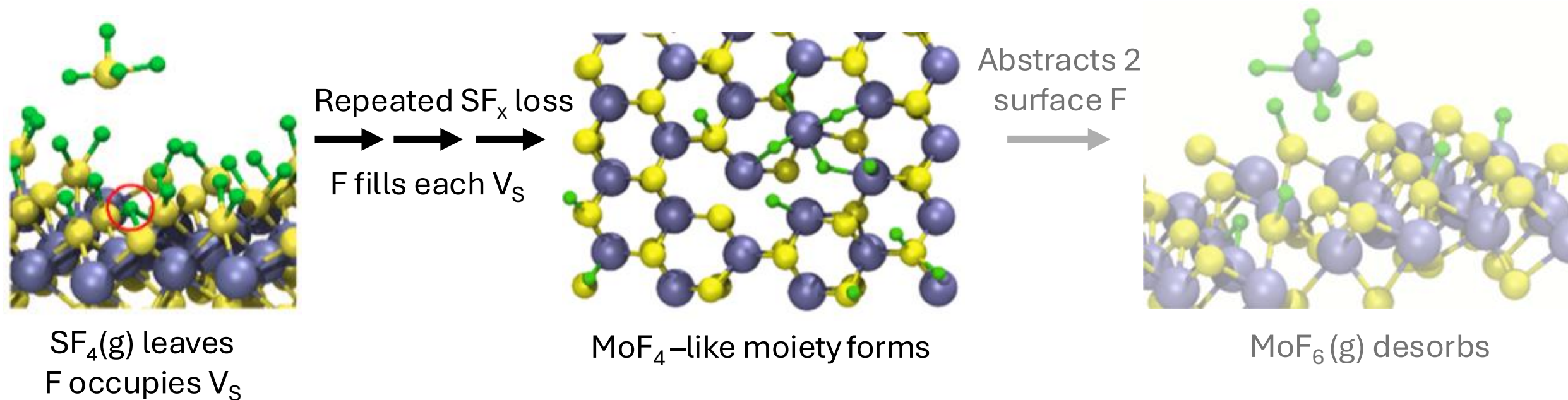
F-filled vacancies promote Mo removal



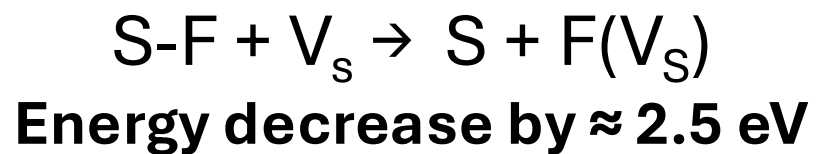
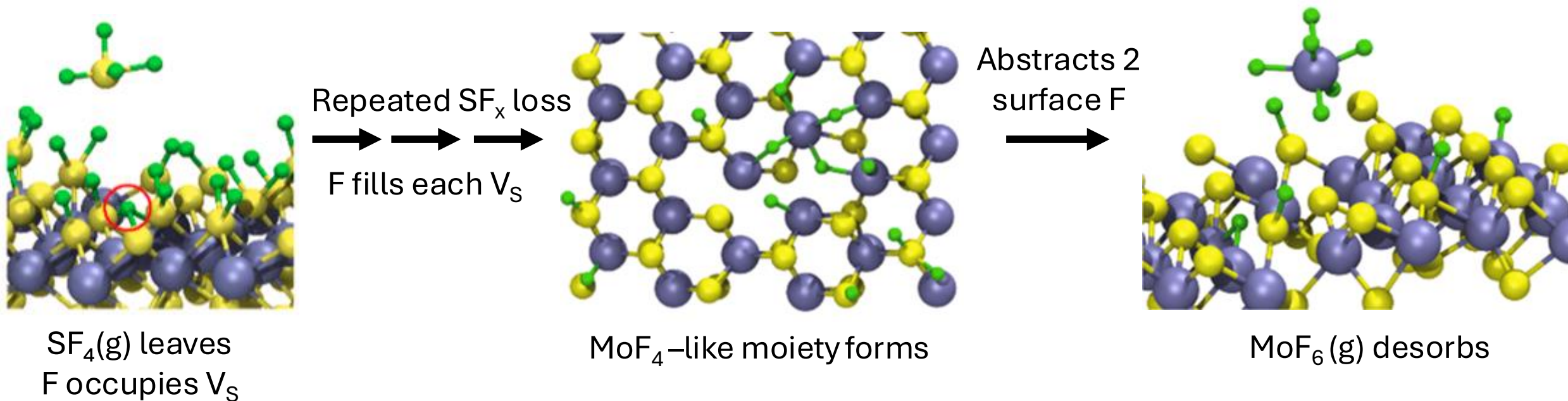
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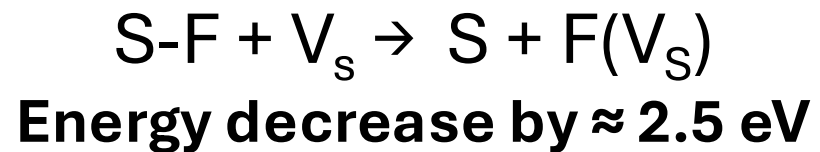
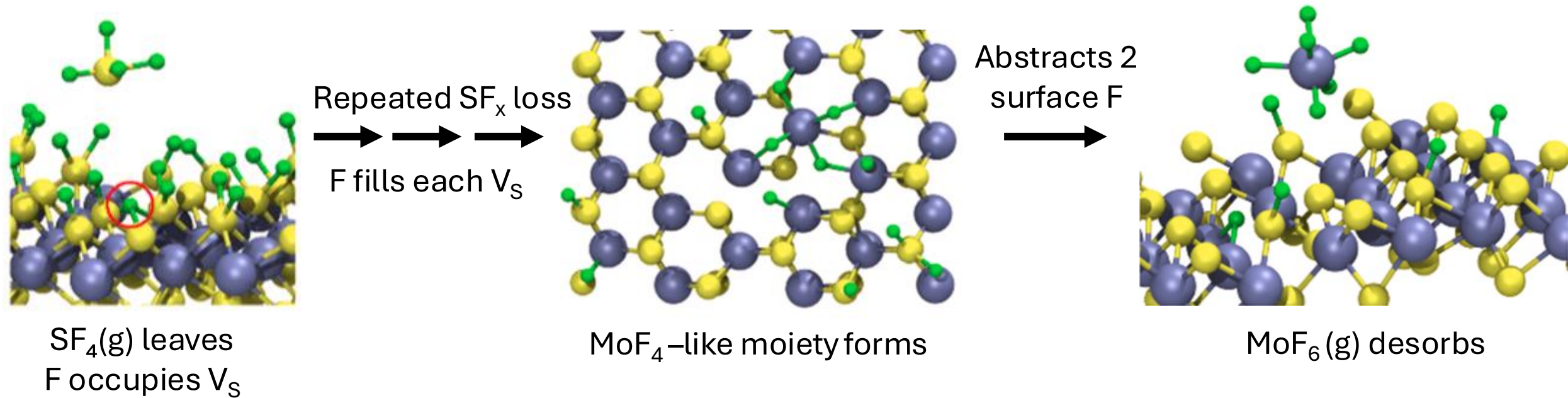
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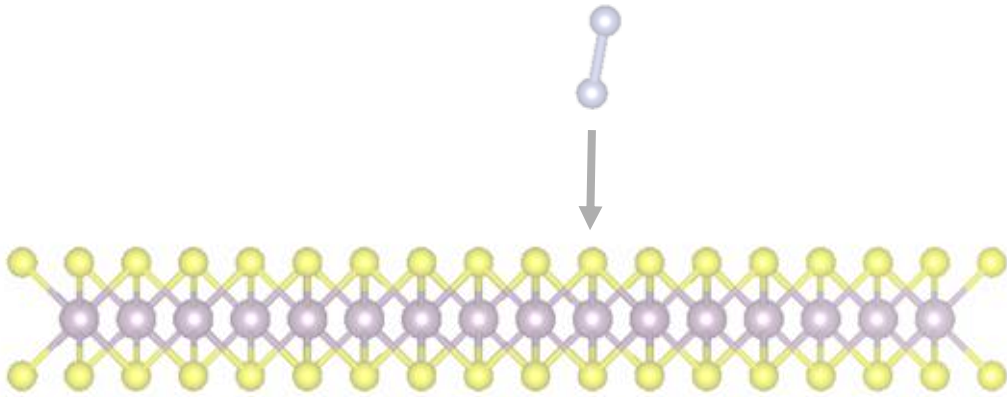
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Can this reactivity be controlled?

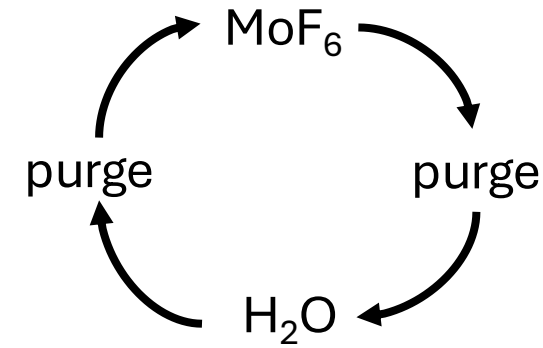
Fluorination mechanisms and etch control

$F_2 + MoS_2 \rightarrow$ atomistic pathway



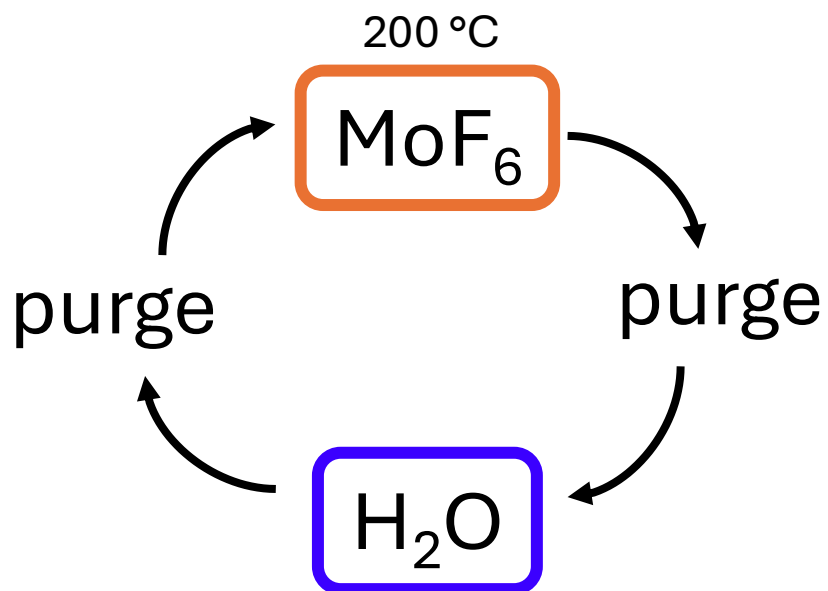
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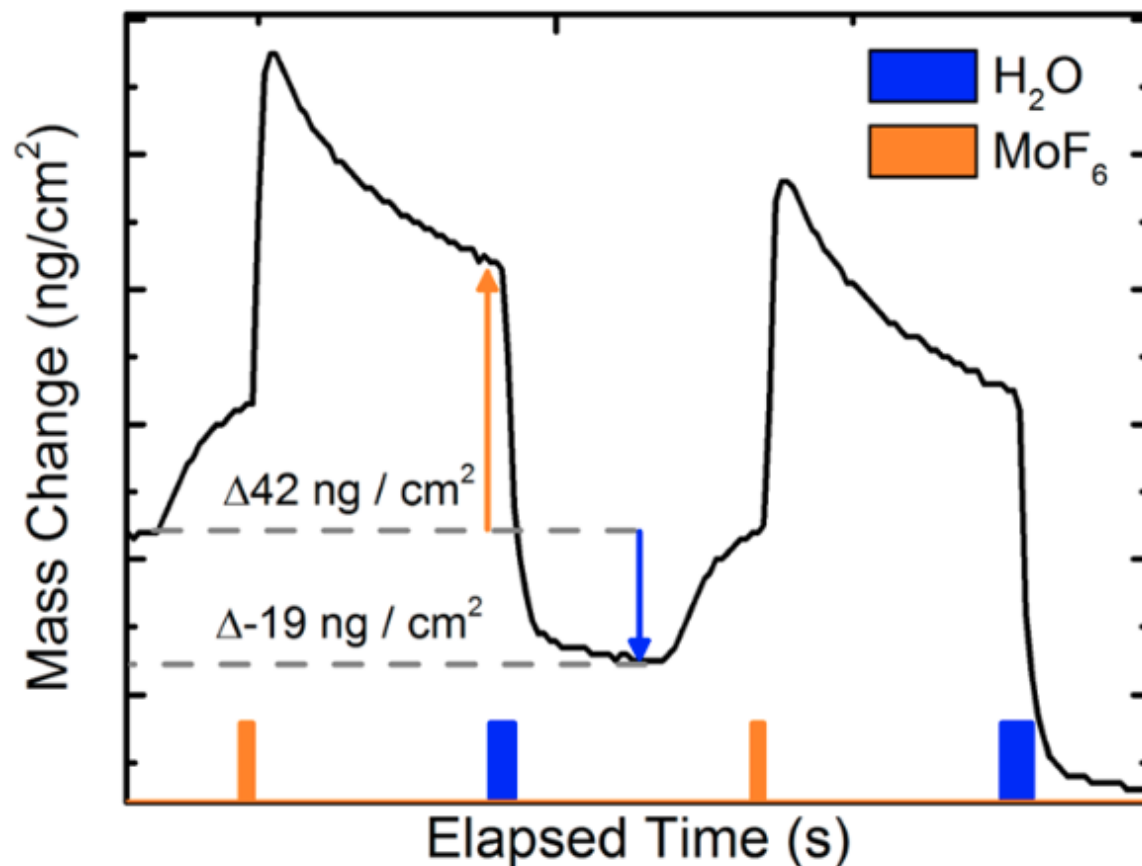


- Can cyclic exposures produce self-limiting removal?
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- Saturating half-reactions and alternating O/F surface states

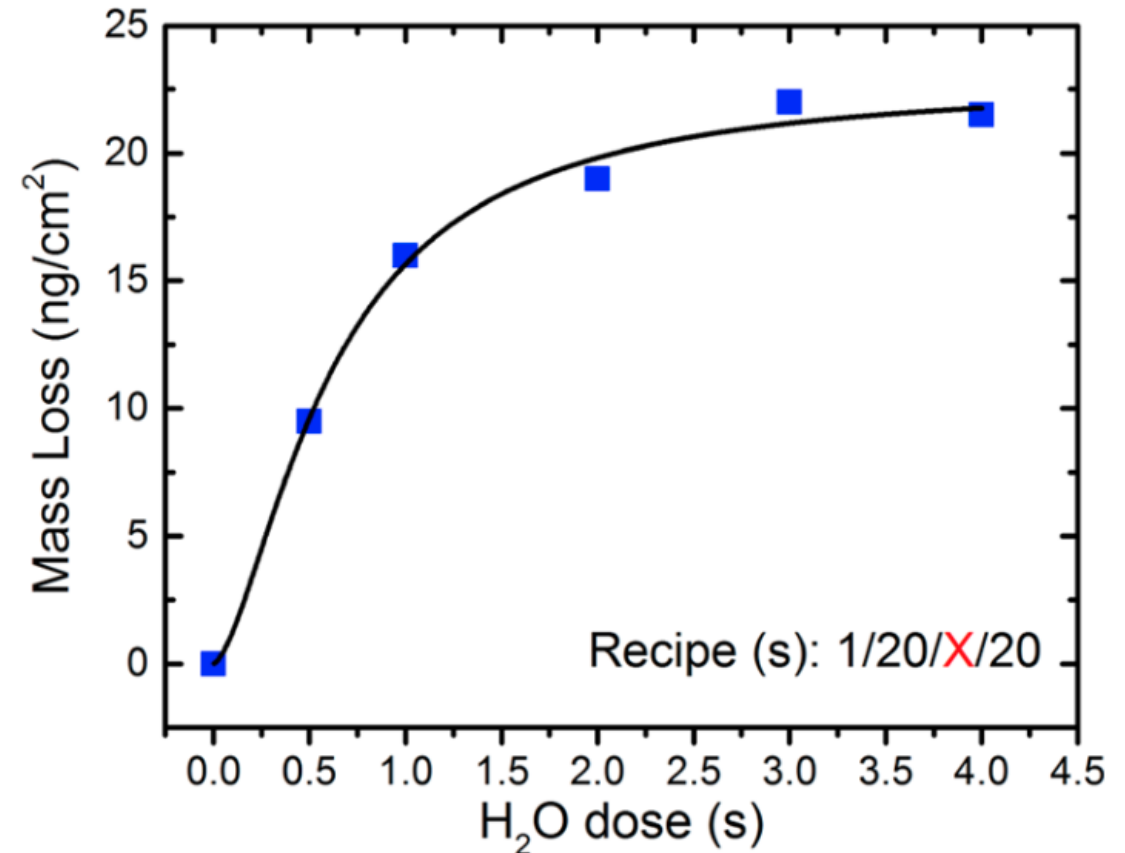
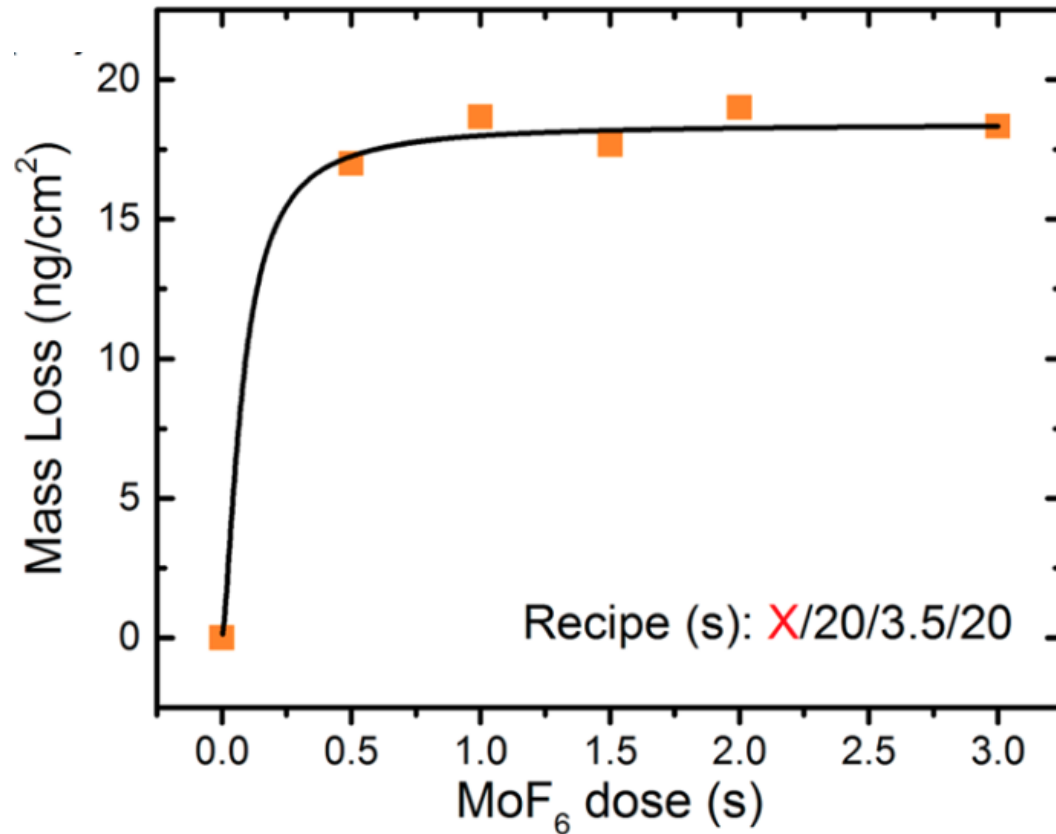
Separating surface modification from removal



- MoF_6 exposure modifies the surface
- H_2O exposure removes surface material
- Complete cycle produces net removal

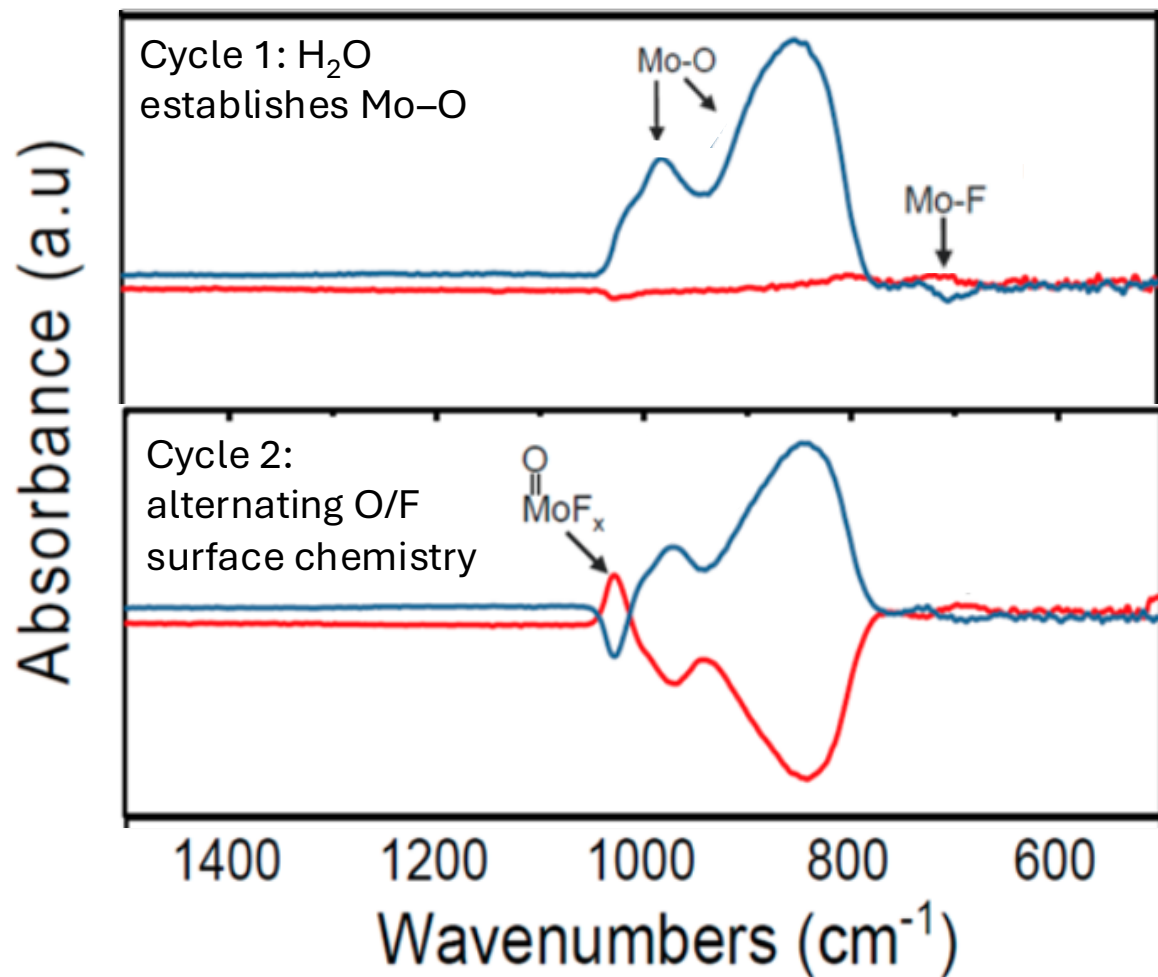


Both half-reactions saturate



- Once the available surface sites are consumed, additional precursor exposure does not increase removal per cycle.

Alternating O/F surface chemistry



- **MoF_6 exposure:**

Mo-O groups consumed
Mo-F, MoF_xO groups form

- **H_2O exposure:**

F-containing groups consumed
Mo-O groups reform

Each exposure prepares the surface for the next half-cycle.

H₂O exposure releases volatile products

Half-cycle	Δm (QCM)	Surface response (FTIR)	Gas-phase evidence (QMS)
MoF ₆ exposure	+42 ng cm ⁻²	Mo-O ↓ Mo-F, MoF _x =O ↑	Not measured
H ₂ O exposure	-61 ng cm ⁻²	Mo-F, MoF _x =O ↓ Mo-O ↑	MoF ₂ O ₂ , H ₂ S, HF, H ₂

Proposed overall reaction:



Continuous vs. self limiting etching

Continuous F₂:

F exposure

S fluorination → SF_x loss → defects → MoF_x loss

ETCH
PROPAGATION

Continuous vs. self limiting etching

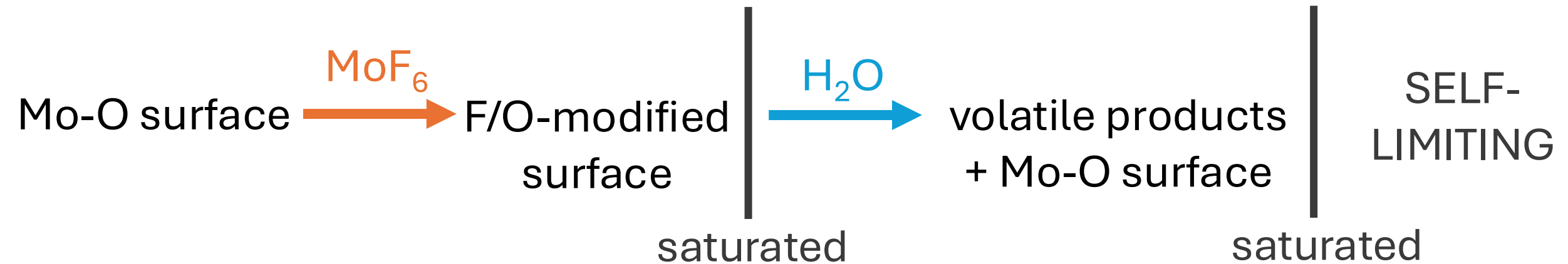
Continuous F_2 :

F exposure

S fluorination $\rightarrow$ SF_x loss $\rightarrow$ defects $\rightarrow$ MoF_x loss

ETCH
PROPAGATION

MoF_6/H_2O cycling:

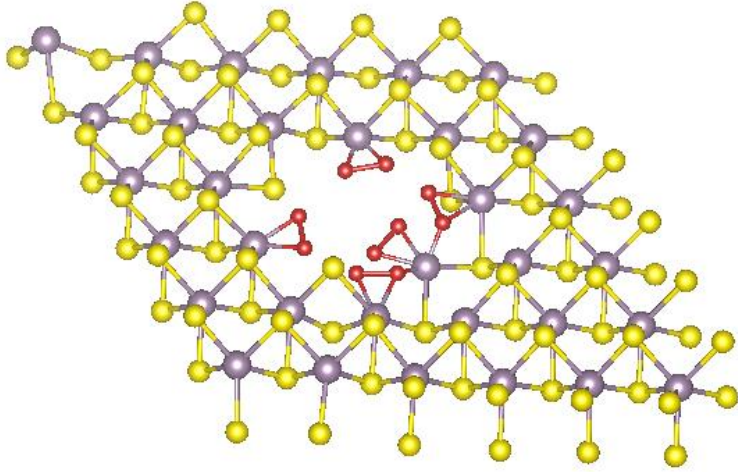


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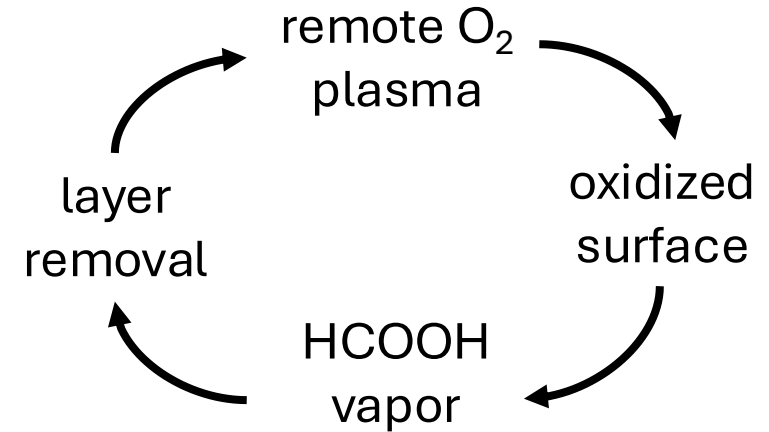
Oxidation mechanisms and etch control

$\text{O}_2 + \text{MoS}_2 \rightarrow$ atomistic pathway



- How does oxidation initiate and propagate?
- DFT, AIMD, Atomic Force Microscopy (AFM), Scanning Electron Microscopy (SEM)
- O incorporation $\rightarrow$ SO_x loss $\rightarrow$ $\text{Mo}_x\text{S}_y\text{O}_z/\text{MoO}_x$ products

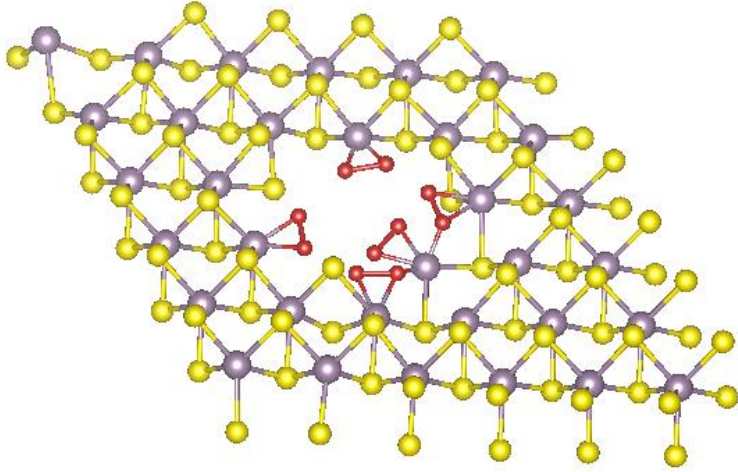
$\text{O}\cdot/\text{HCOOH} \rightarrow$ controlled removal



- Can oxidation and removal be separated to achieve layer-by-layer etching?
- X-ray Photoelectron Spectroscopy (XPS), Raman Spectroscopy, AFM
- Alternating precursor exposure, layer by layer removal

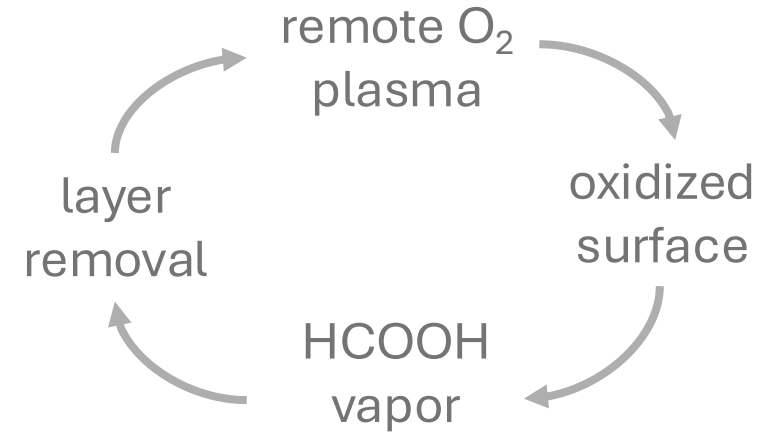
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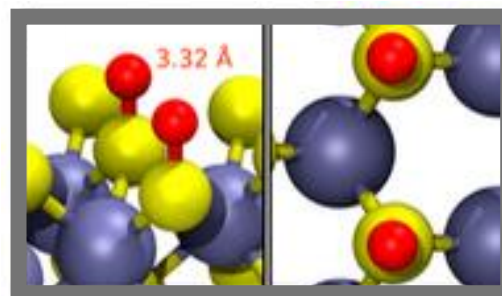
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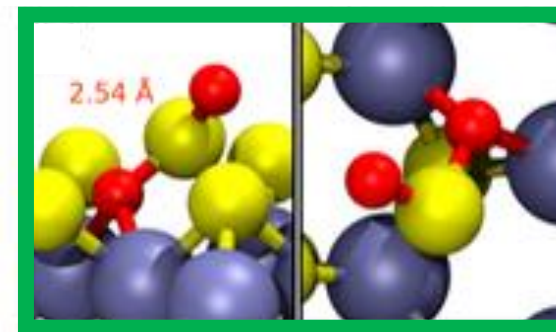
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Multiple intermediates in MoS₂ oxidation

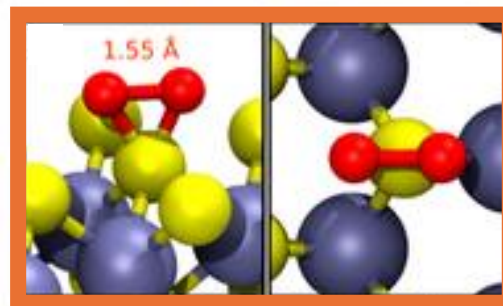
- Five local minima along candidate SO_x-desorption pathways
- AIMD tested whether each intermediate persisted, rearranged, or released SO_x



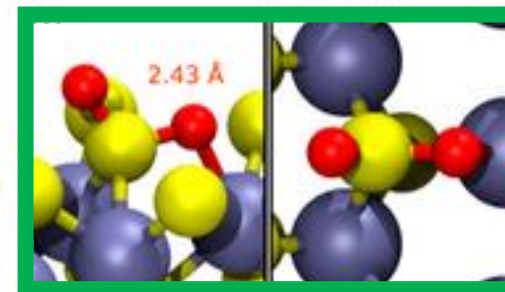
SO+SO $\Delta E = -1.08$ eV



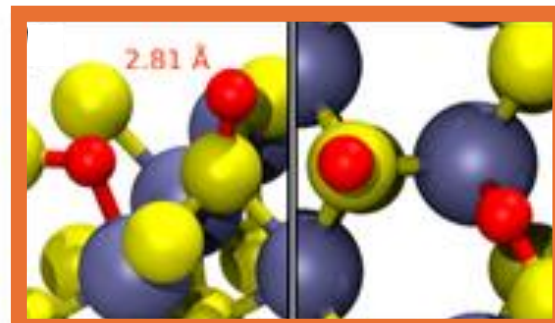
Mo-OSO $\Delta E = -0.54$ eV



O₂-S $\Delta E = 1.87$ eV



OSOMo $\Delta E = -0.71$ eV



SO+SOMo $\Delta E = 0.03$ eV

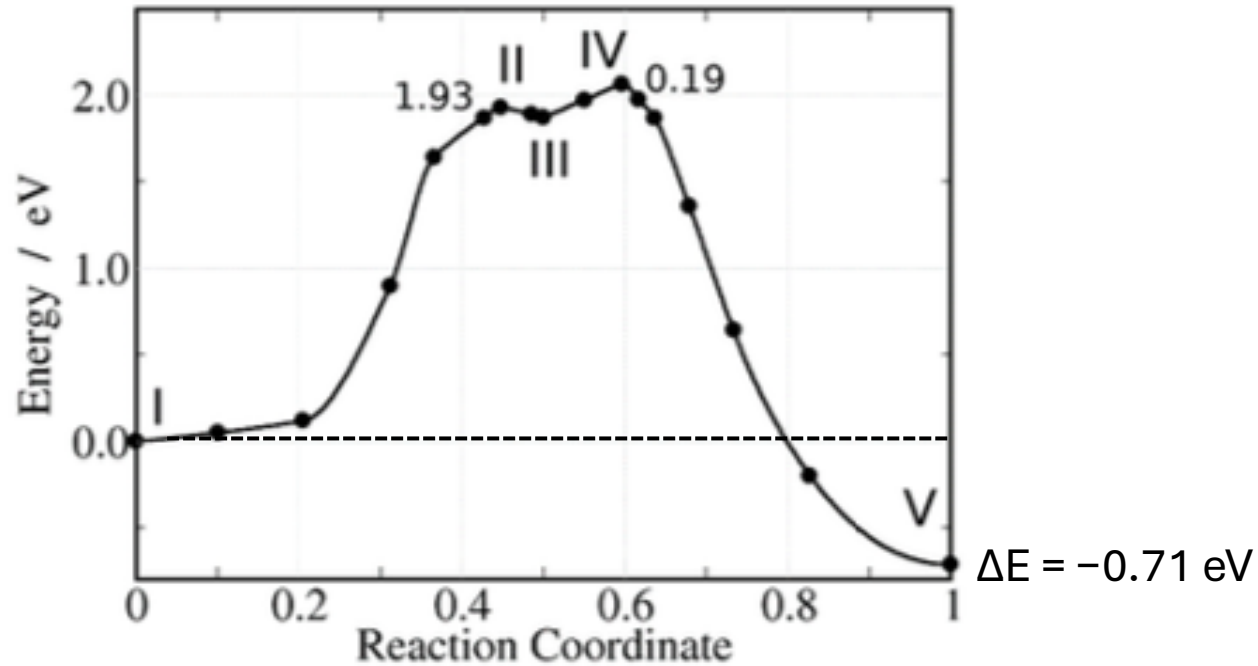
Kinetic trap

Rearranges

Enables SO_x
loss

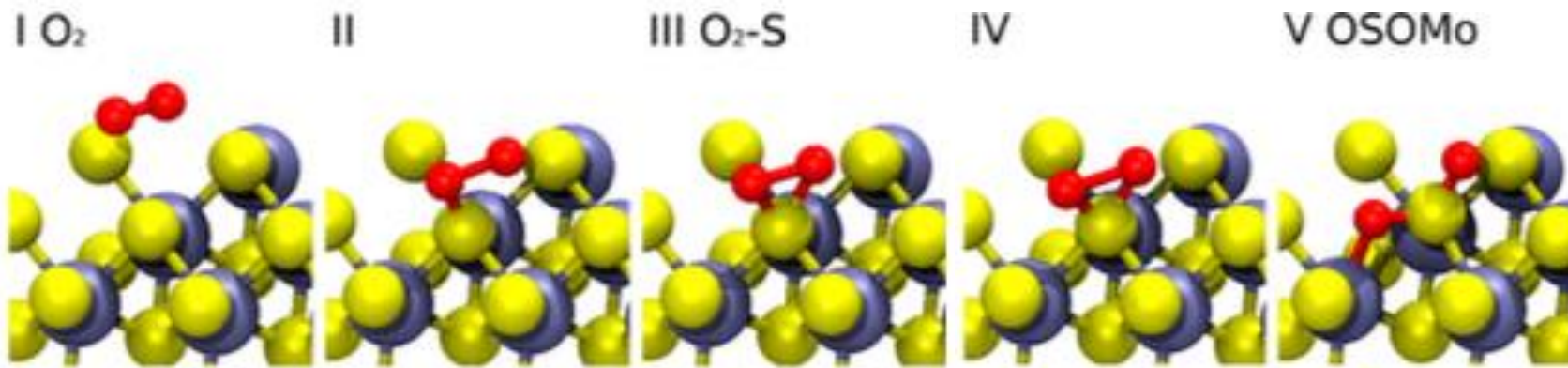
**Different bonding
arrangements produce
different reactivity**

OSOMo formations enables sulfur loss

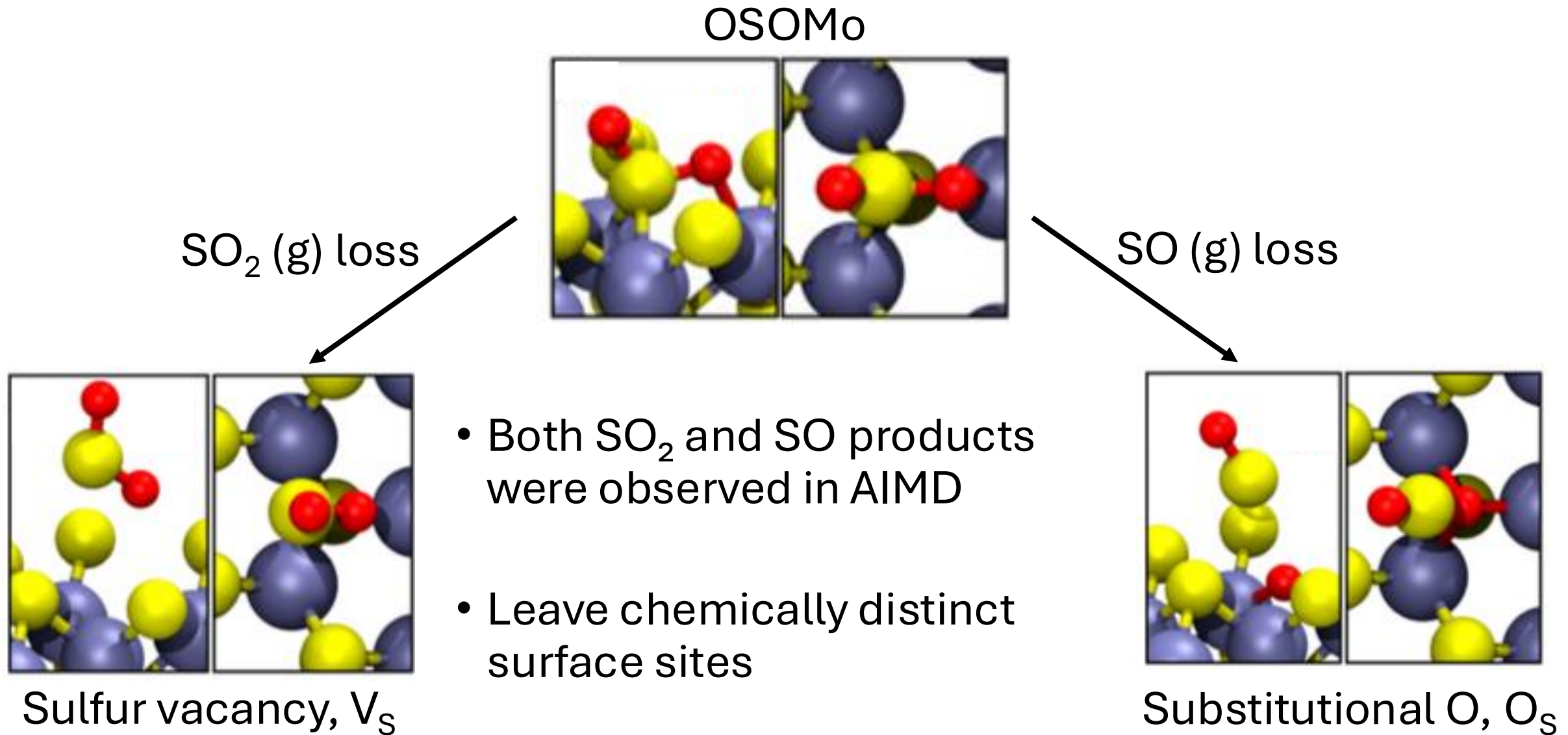


- O_2 activation is difficult on pristine basal MoS_2
- Insertion into S-Mo has a low local barrier after O_2 -S forms

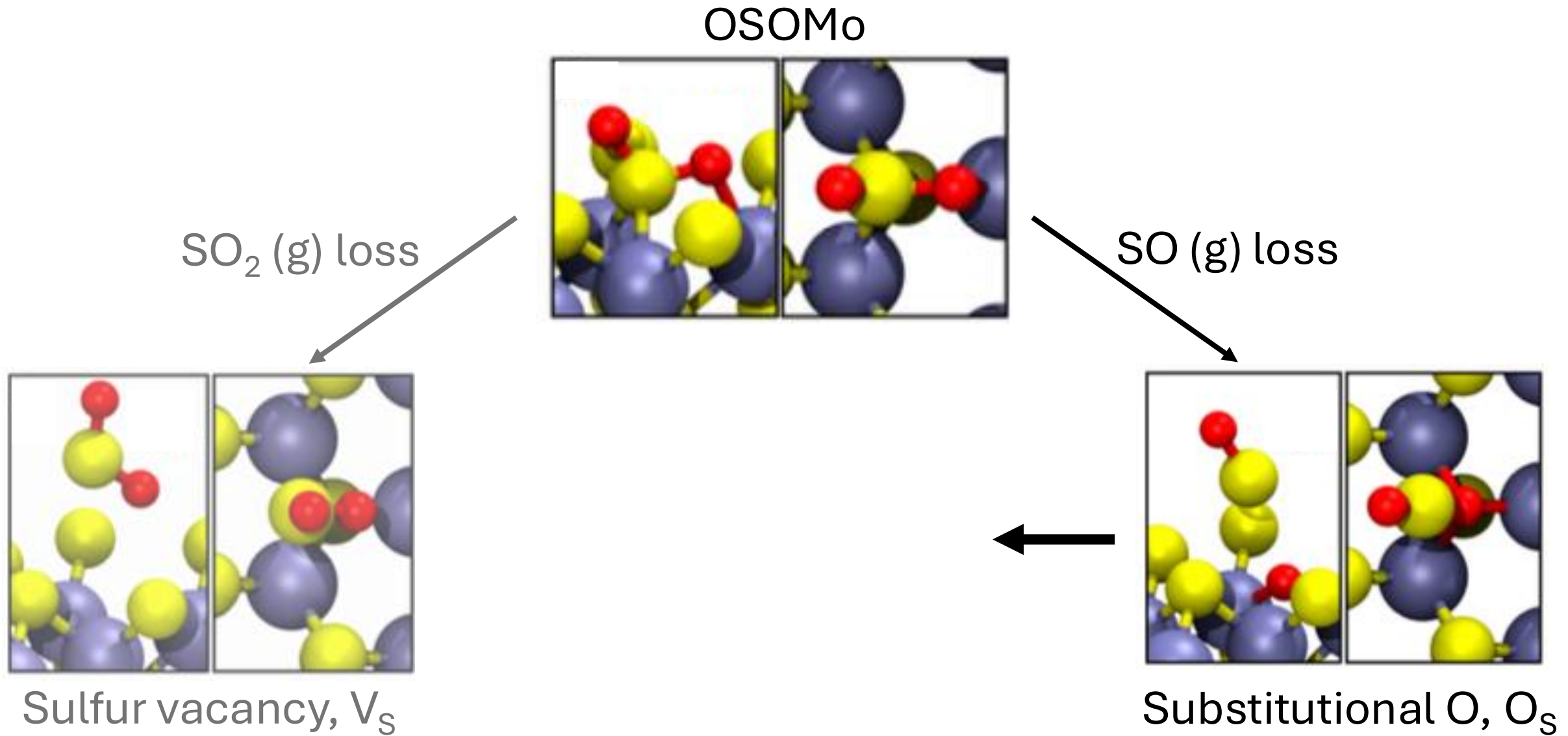
OSOMo enables sulfur loss...



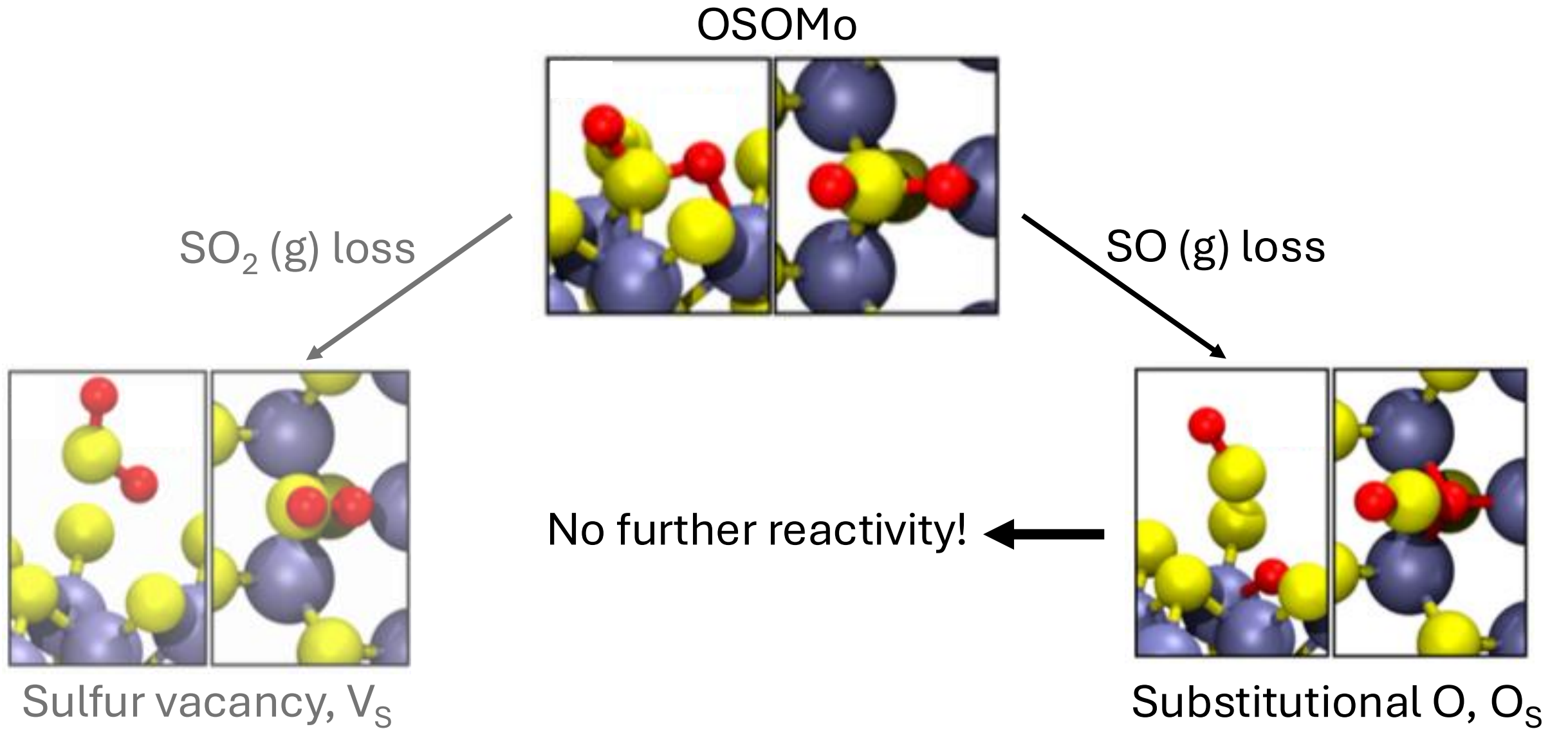
Sulfur loss determines the surface state



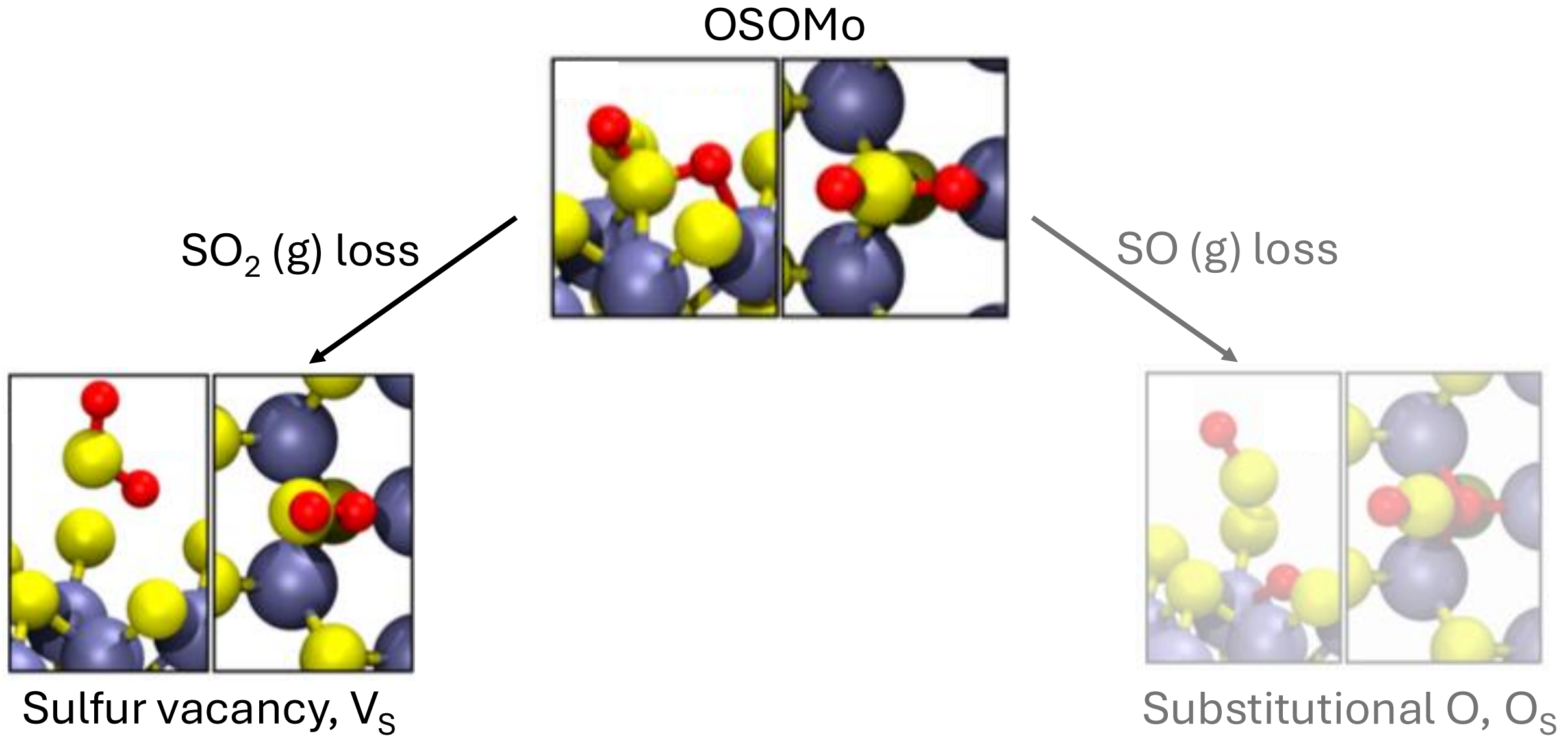
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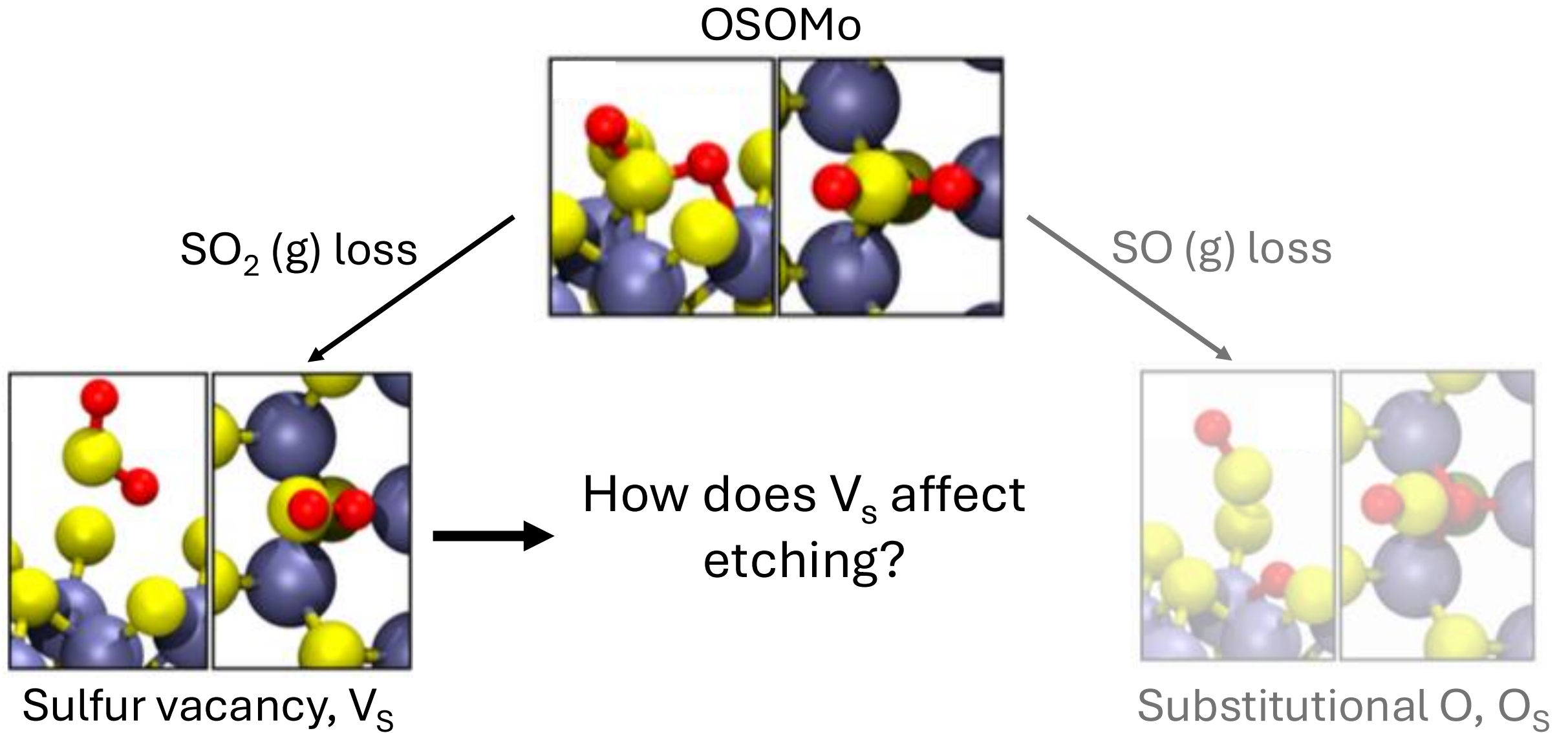
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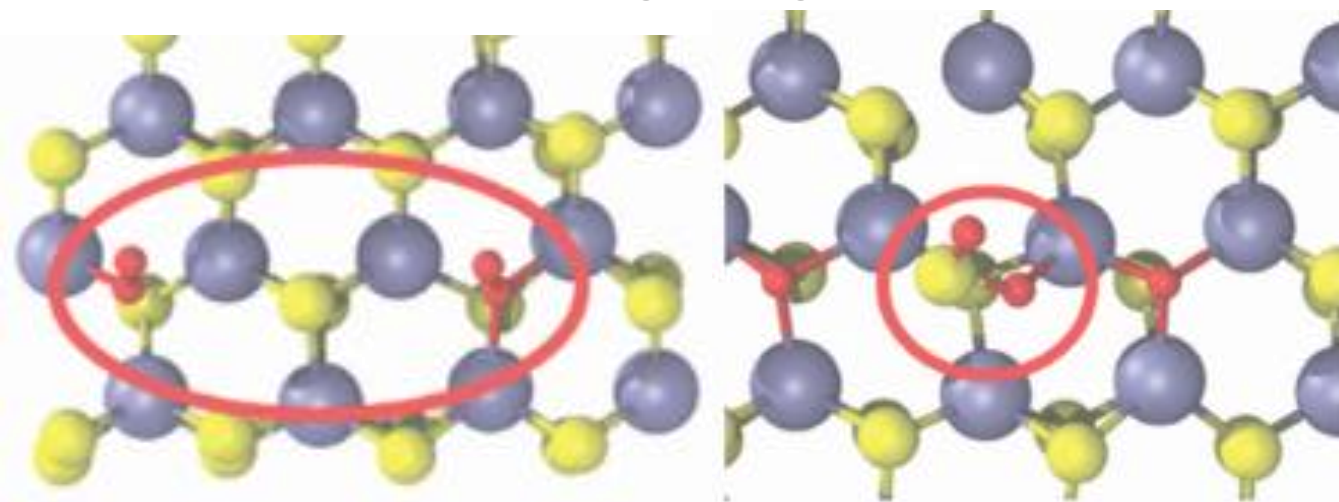
Vacancy geometry matters

Isolated V_s



- O_2 dissociates, one O fills V_s . other forms a nearby S-O/Mo-S-O-Mo group
- No SO/SO₂ desorption observed in AIMD

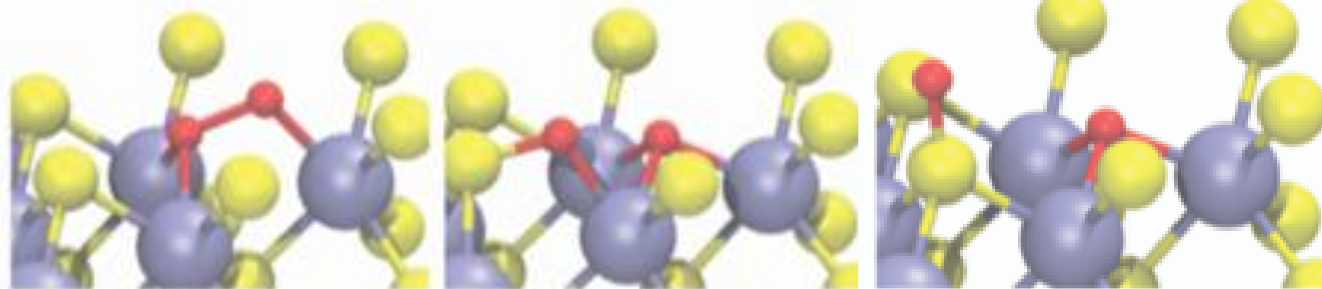
V_s -S- V_s



- Two neighboring O_2 form an O-S-O-Mo intermediate
- SO₂ desorption observed in AIMD

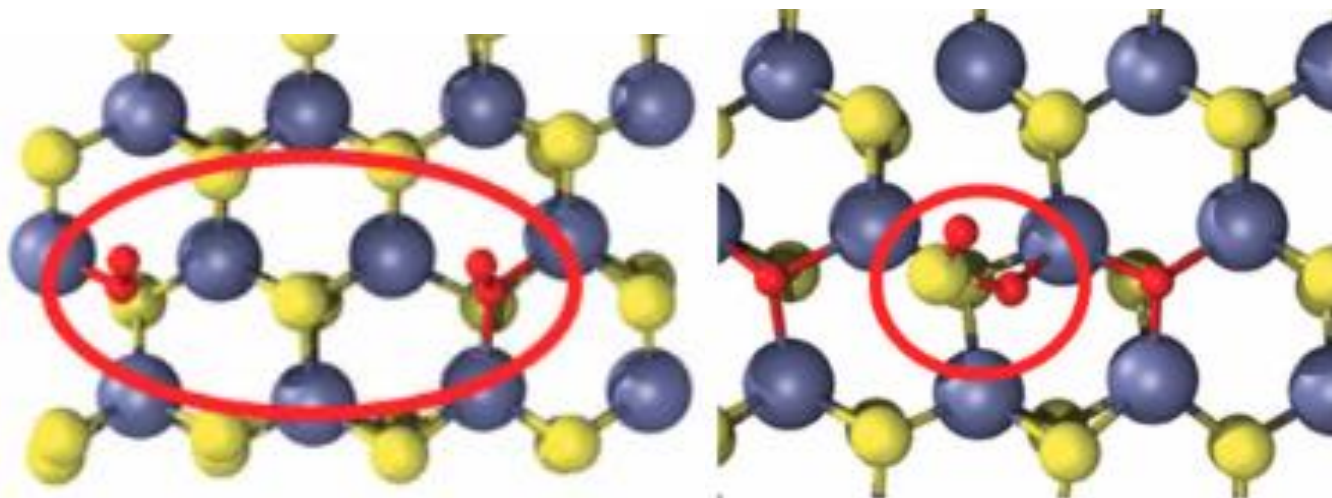
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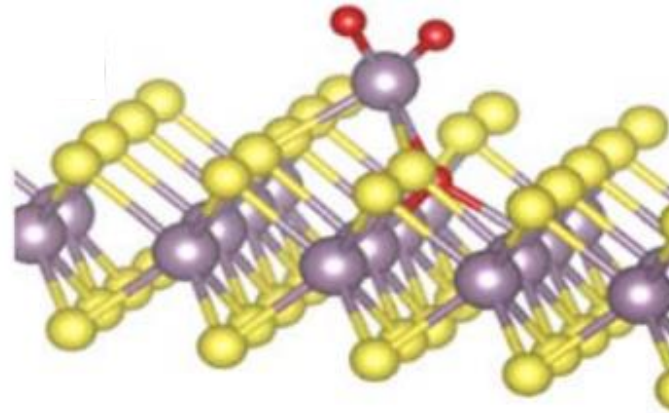
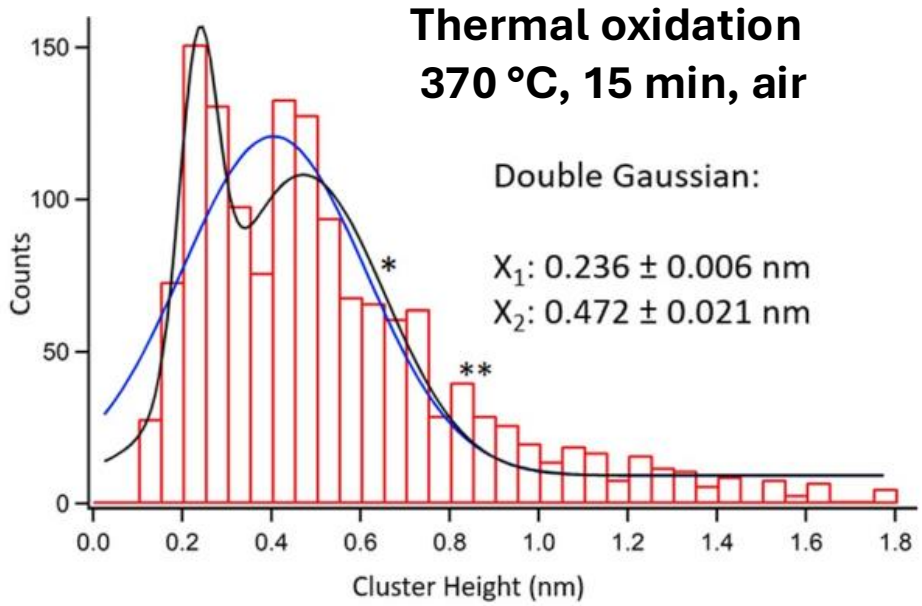
- O_2 dissociates, one O fills V_s . other forms a nearby S-O/Mo–S-O-Mo group
- No SO/SO₂ desorption observed in AIMD

V_s -S- V_s

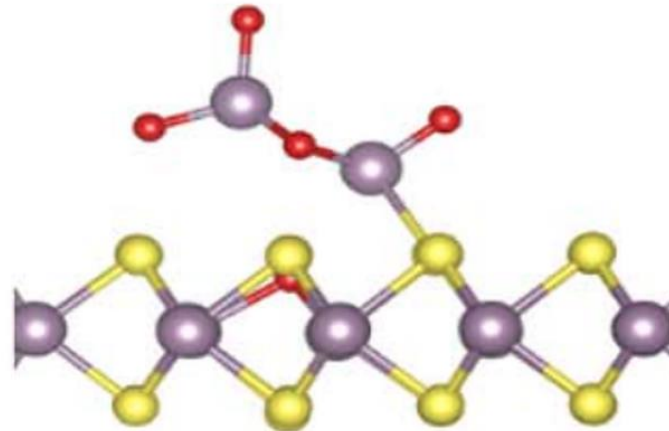


- Two neighboring O_2 form an O-S-O-Mo intermediate
- SO₂ desorption observed in AIMD

MoO₃-like clusters remain after etching

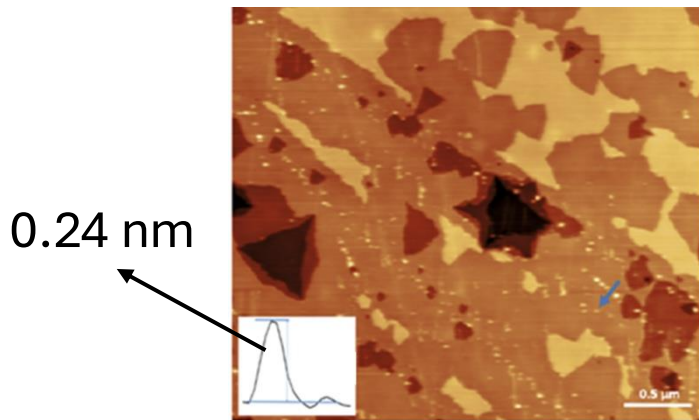


Mo₁O₃/MoS_{2-x}
Height = 0.243 nm



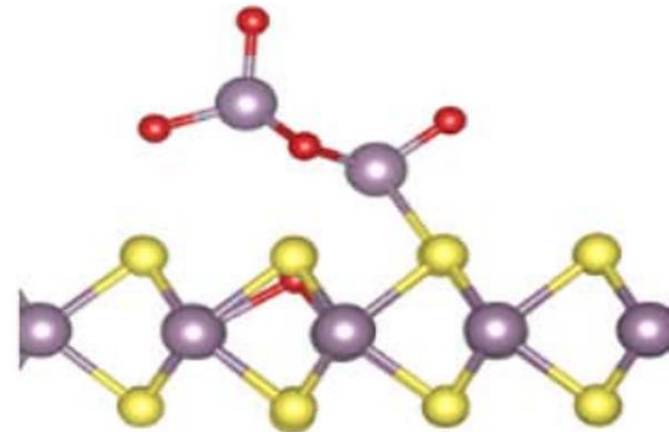
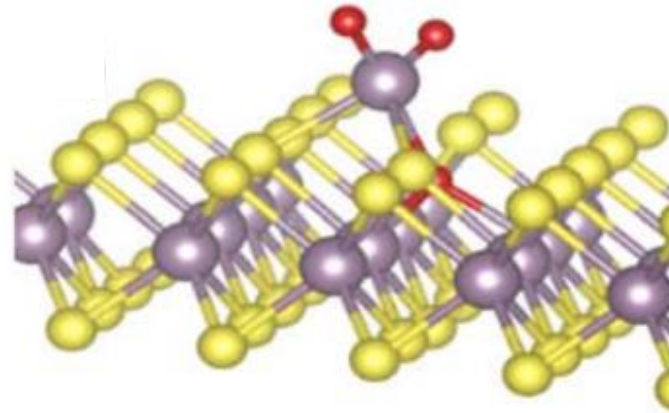
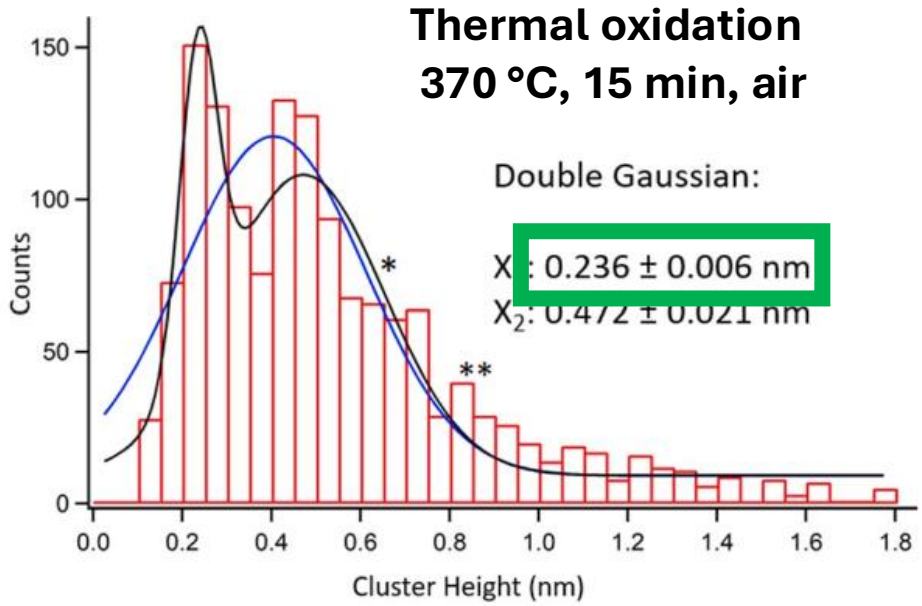
Mo₂O₆/MoS_{2-x}
Height = 0.484 nm

- AFM revealed 2 cluster height populations
- DFT structures matched both heights: MoO₃ and Mo₂O₆
- Other analysis independently supported MoO₃

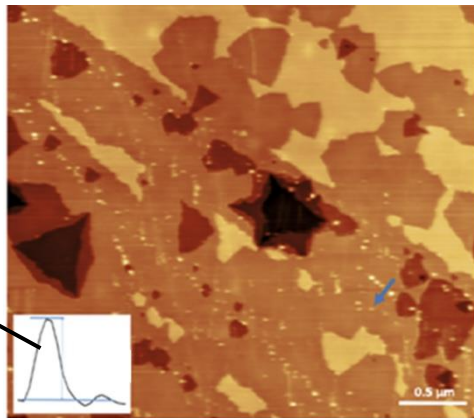


AFM after thermal oxidation

MoO₃-like clusters remain after etching

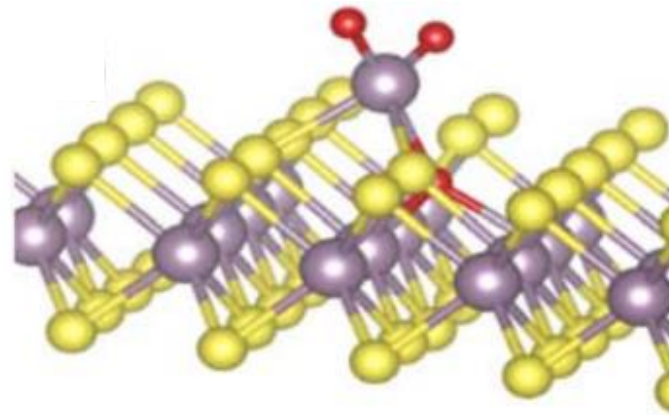
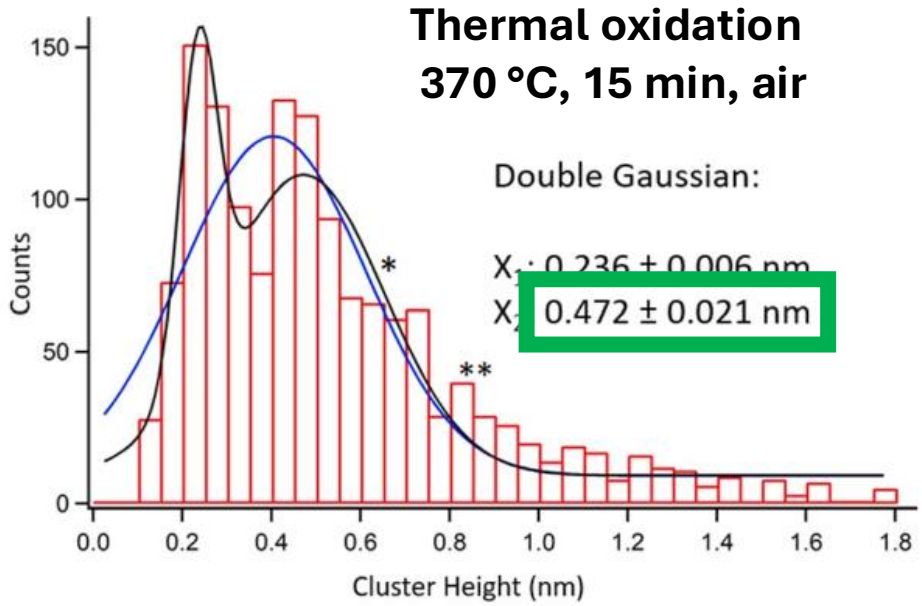


- AFM revealed 2 cluster height populations
- DFT structures matched both heights: MoO₃ and Mo₂O₆
- Raman and X-ray absorption near edge spectroscopy independently supported MoO₃

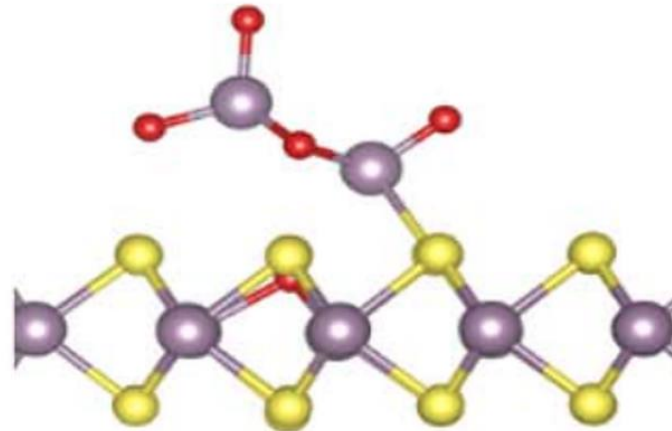


AFM after thermal oxidation

MoO₃-like clusters remain after etching

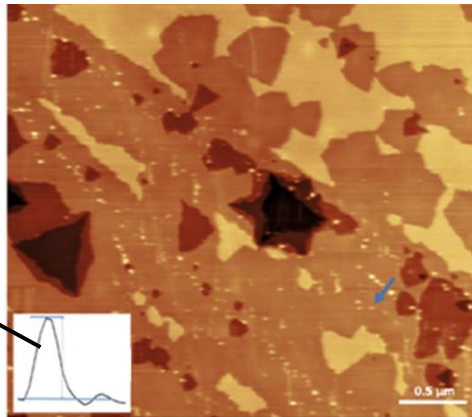


Mo₁O₃/MoS_{2-x}
Height = 0.243 nm



Mo₂O₆/MoS_{2-x}
Height = 0.484 nm

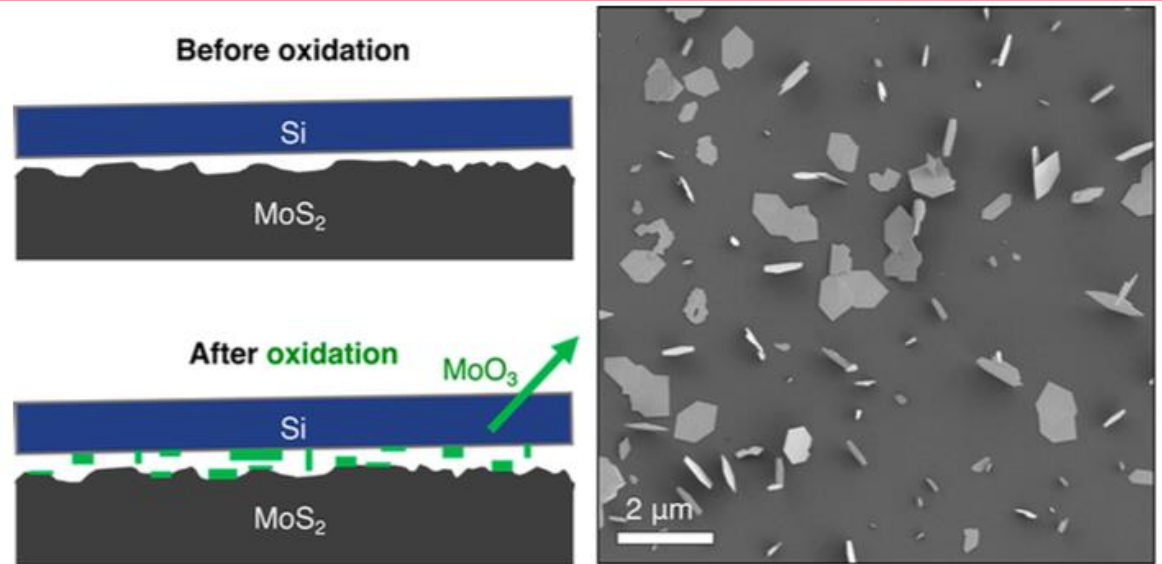
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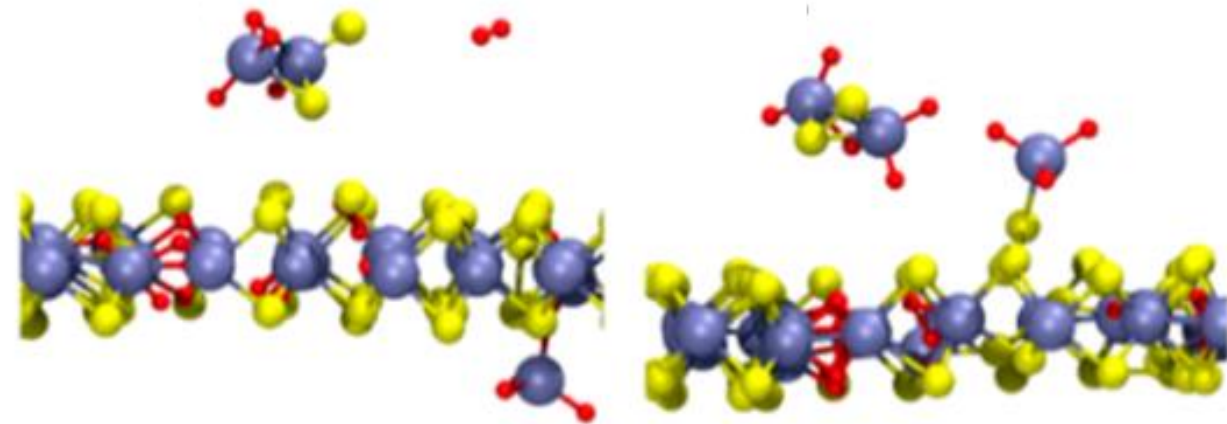
AFM after thermal oxidation

Volatile Mo-containing products

- Separate study shows as MoO_3 form across a micrometer gap
- AIMD predicts mixed $\text{Mo}_x\text{S}_y\text{O}_z$ complexes can detach from oxidized pit edges



500 °C

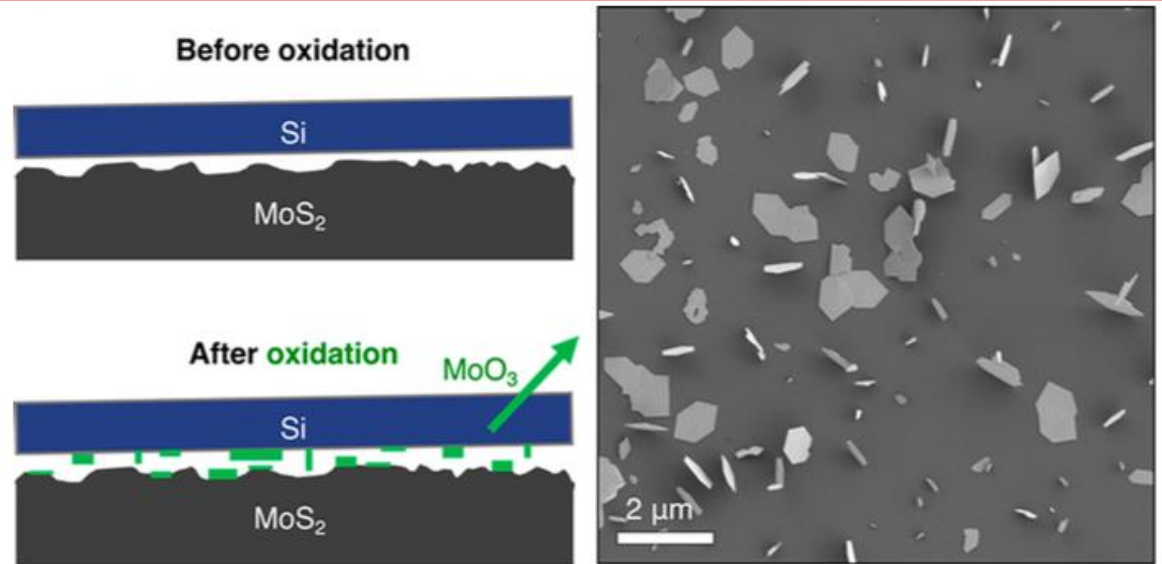


1500 K

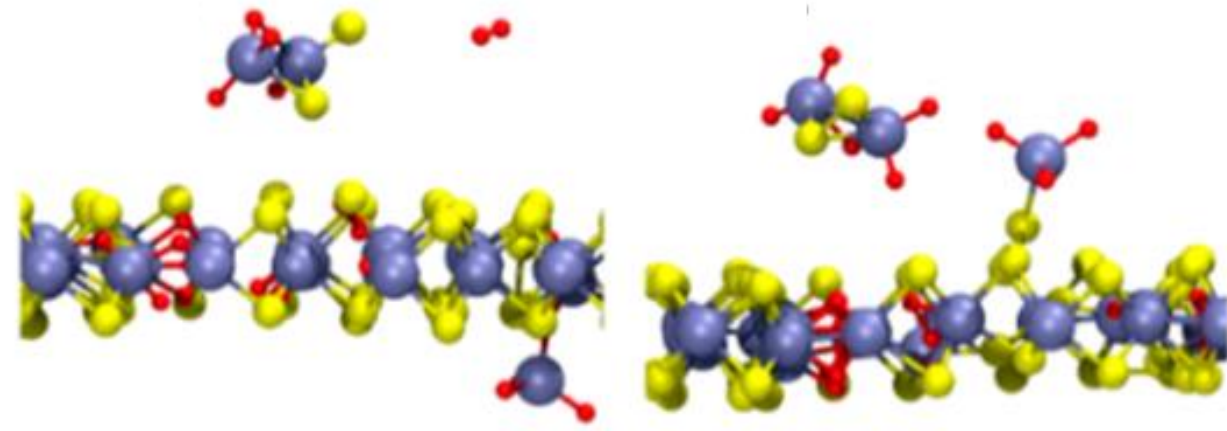
Volatile Mo-containing products

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**Can oxidation and removal
be separated to enable
cyclic etching?**



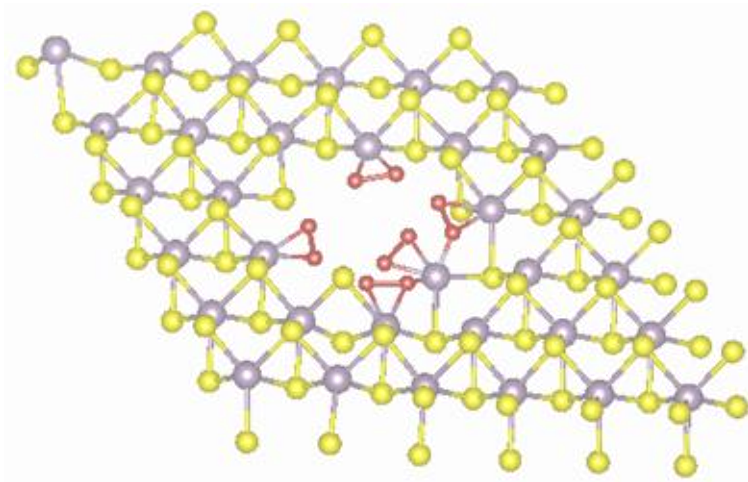
500 °C



1500 K

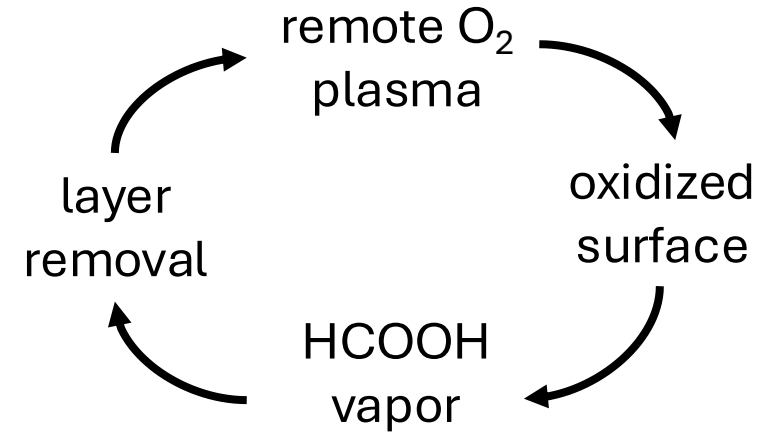
Oxidation mechanisms and etch control

$\text{O}_2 + \text{MoS}_2 \rightarrow$ atomistic pathway



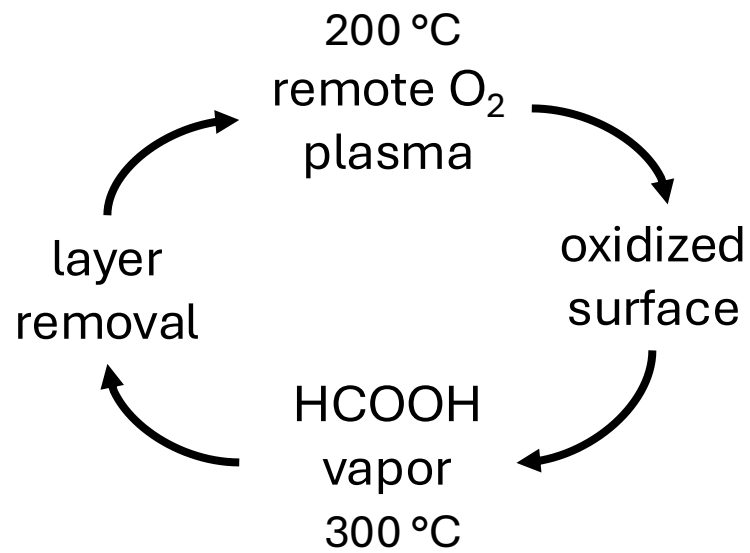
- How does oxidation initiate and propagate?
- DFT, AIMD, Atomic Force Microscopy (AFM), Scanning Electron Microscopy (SEM)
- O incorporation $\rightarrow$ SO_x loss $\rightarrow$ $\text{Mo}_x\text{S}_y\text{O}_z/\text{MoO}_x$ products

$\text{O}\cdot/\text{HCOOH} \rightarrow$ controlled removal

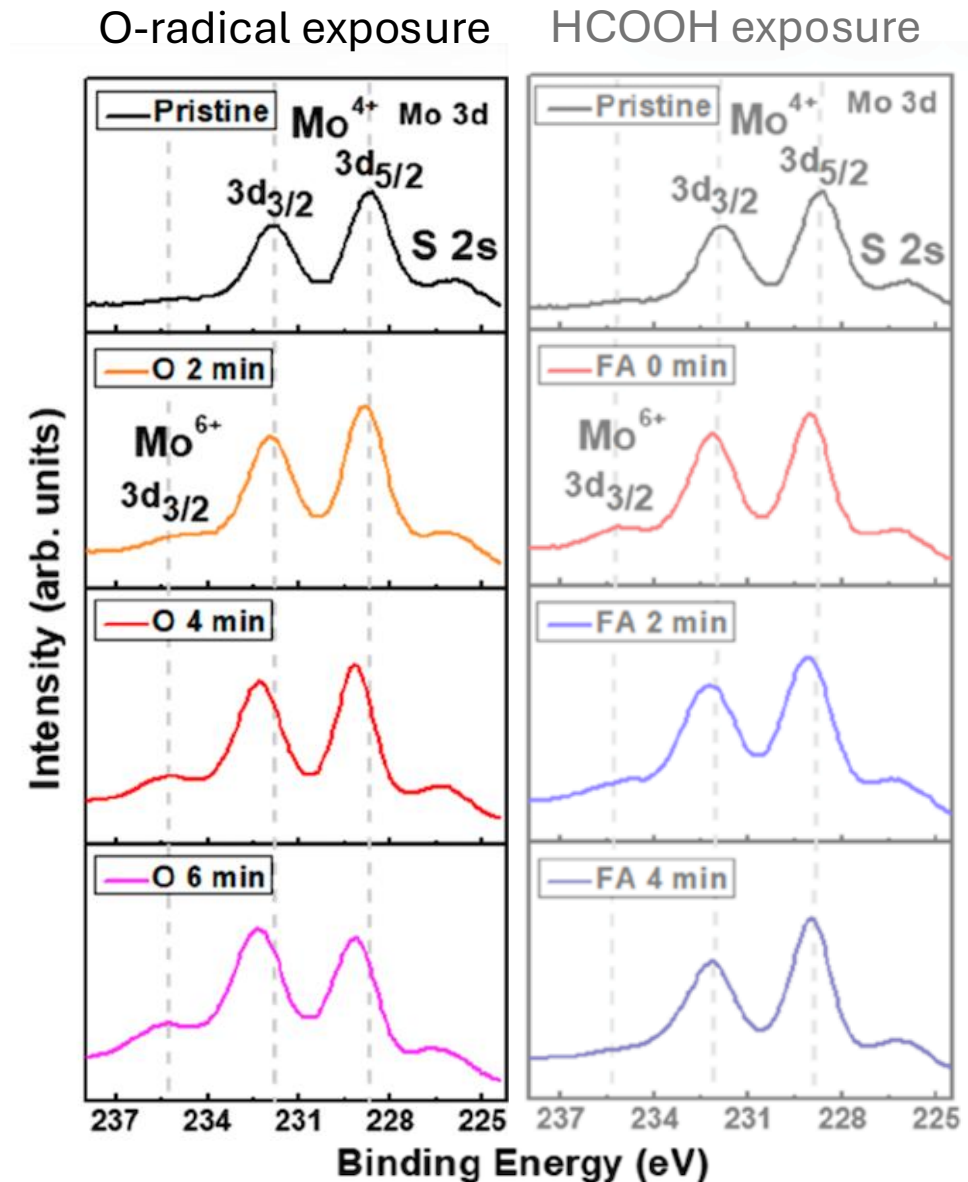


- Can oxidation and removal be separated to achieve layer-by-layer etching?
- X-ray Photoelectron Spectroscopy (XPS), Raman Spectroscopy, AFM
- Alternating precursor exposure, layer by layer removal

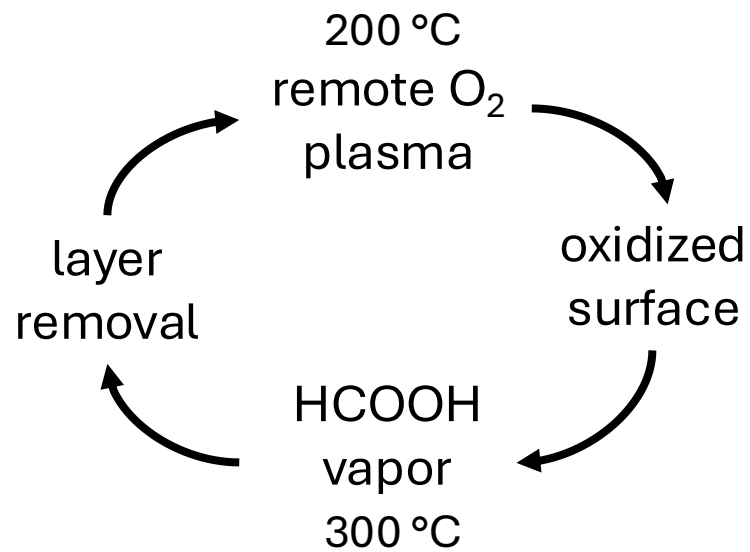
Separating oxidation from removal



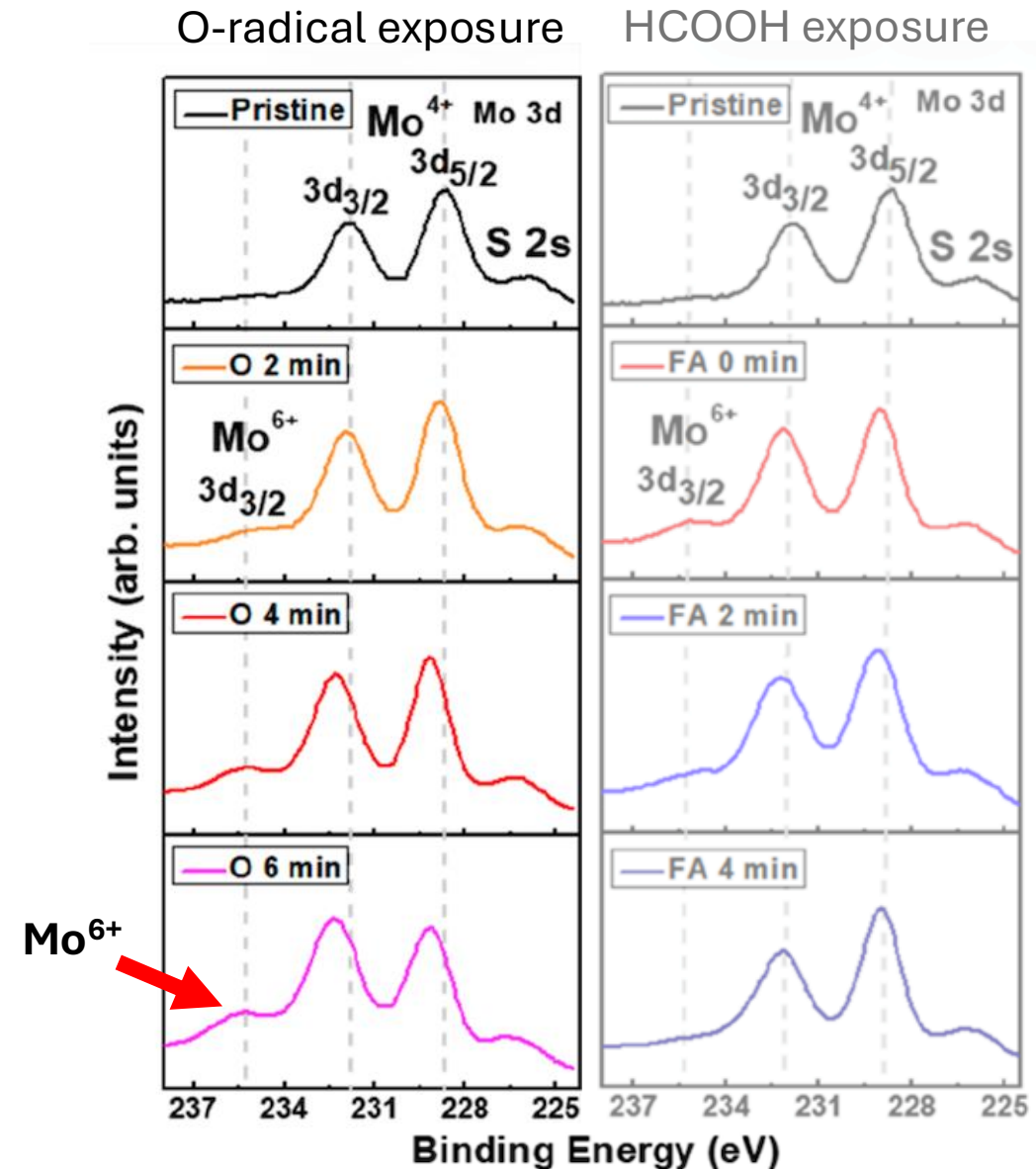
- O-radical exposure forms Mo(VI)-containing MoS_xO_y surface
- HCOOH vapor removes oxidized surface layer
- Complete cycle produces controlled thinning: 3L $\rightarrow$ 2L $\rightarrow$ 1L



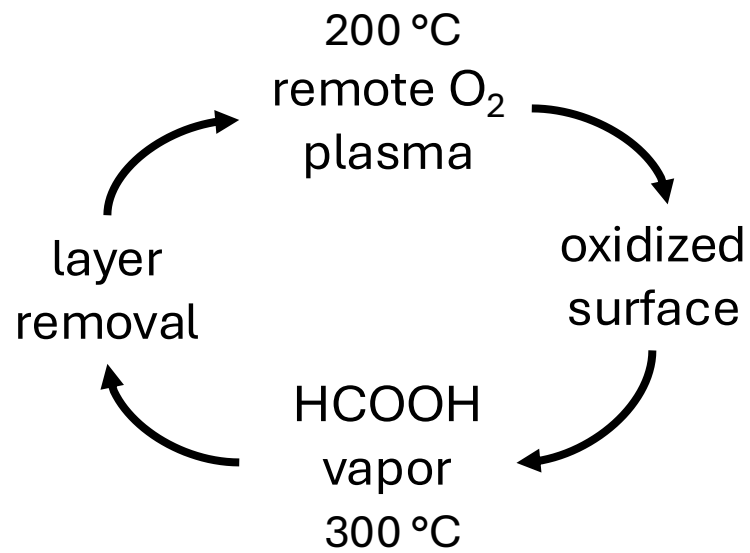
Separating oxidation from removal



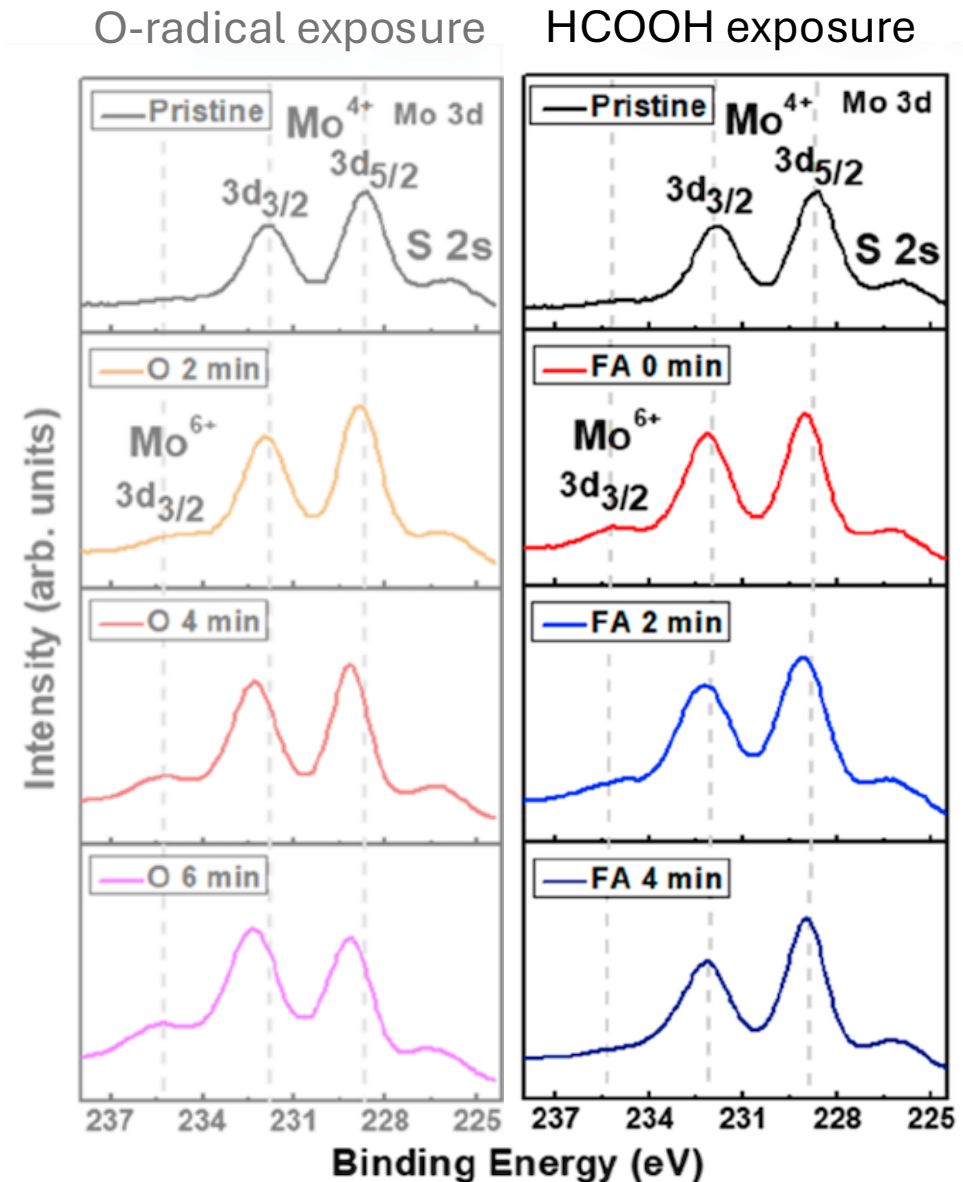
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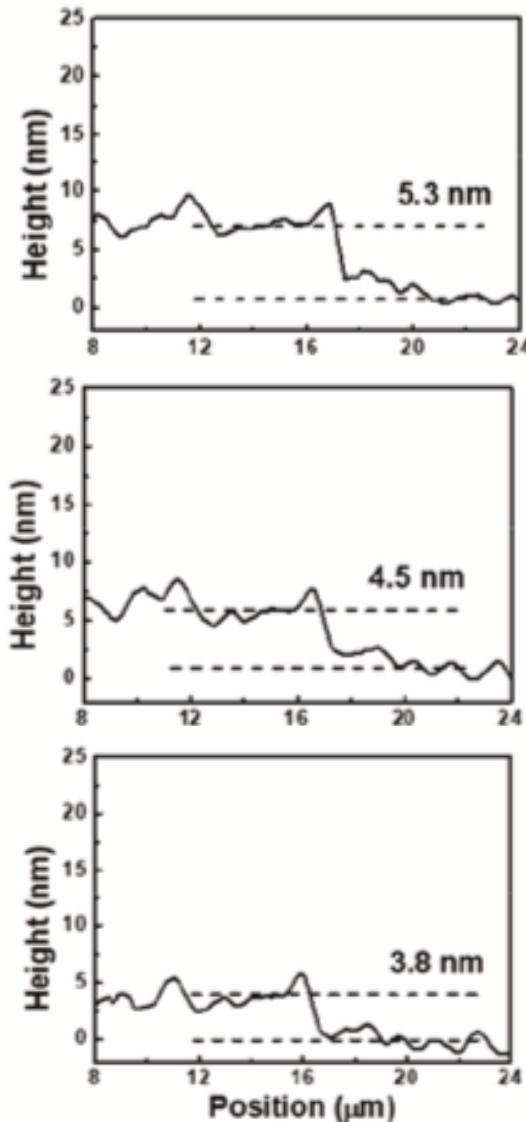
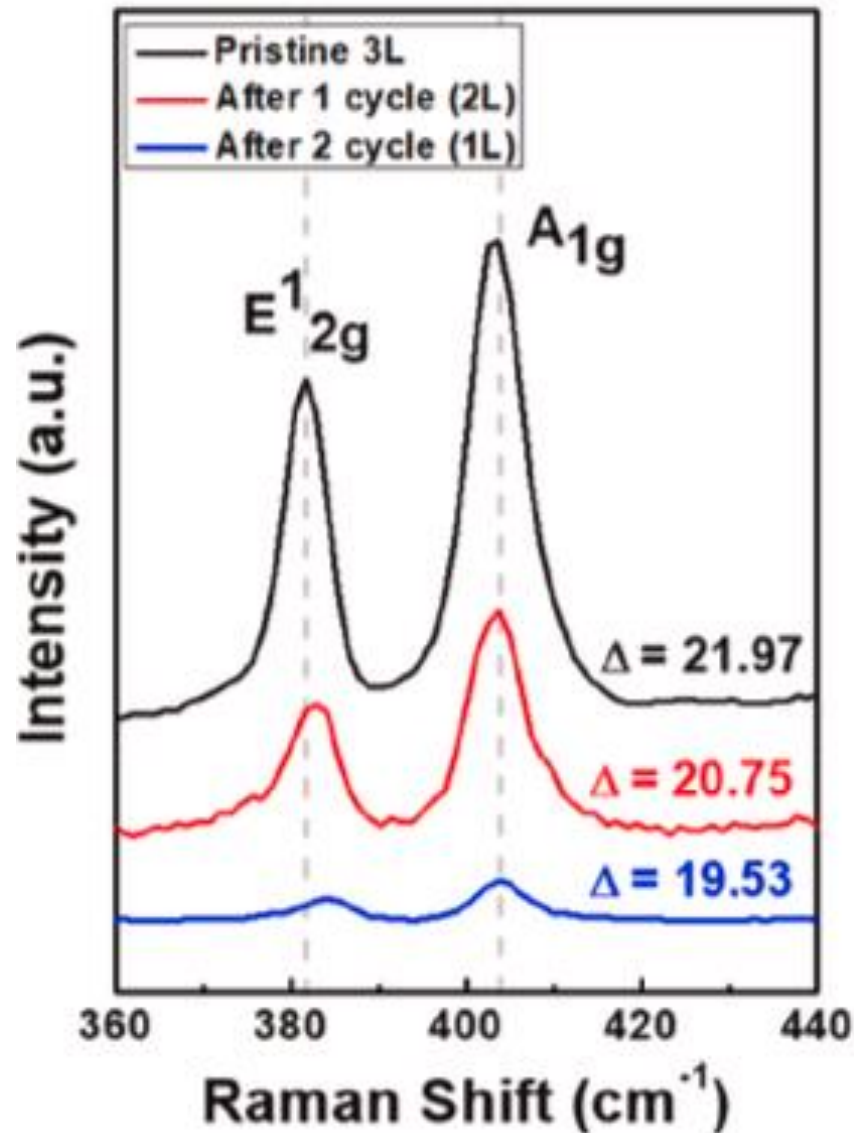
Separating oxidation from removal



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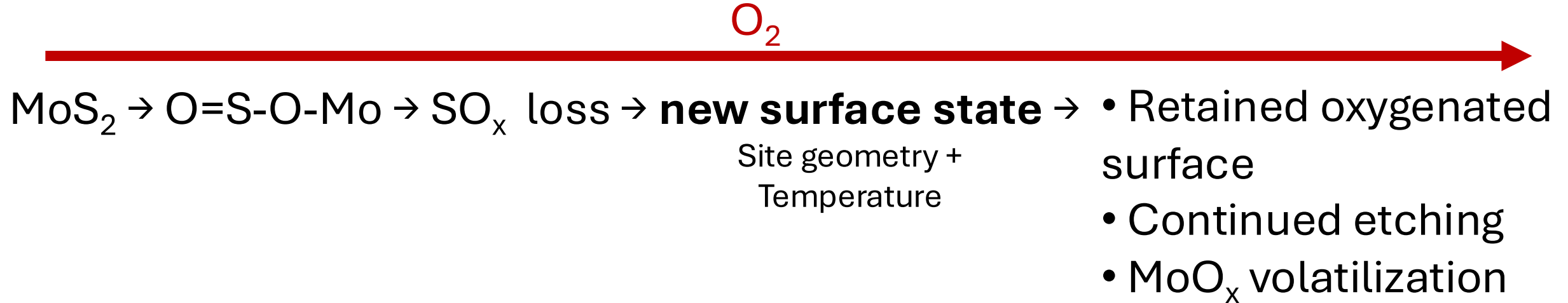
Each cycle removes approximately one layer



- Raman mode separation corresponds to 3L $\rightarrow$ 2L $\rightarrow$ 1L
- AFM height decreases ≈ 0.7 – 0.8 nm/cycle
- Cycle-controlled thinning is demonstrated

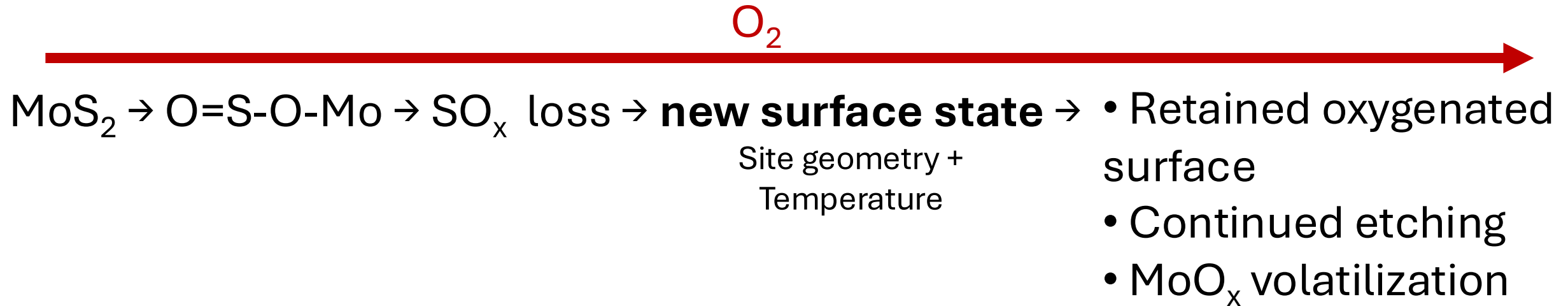
Process conditions redirect oxidation

Continuous thermal oxidation:

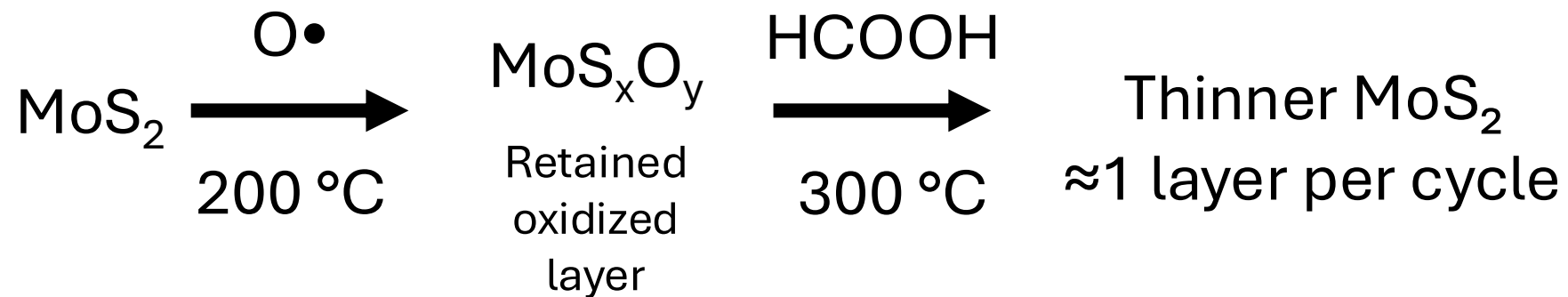


Process conditions redirect oxidation

Continuous thermal oxidation:



$O\bullet$ / $HCOOH$ cycling:



Outline

- Background
 - Material structure and applications
 - Etching process and control
 - Fluorination vs. oxidation
- Fluorination
 - Atomistic Pathway
 - Controlled Etching
- Oxidation
 - Atomistic Pathway
 - Controlled Etching
- **Summary/Outlook**

Overview of mechanisms

Fluorination

Mechanism:

- F_2 activates readily
- SF_x loss and vacancy growth propagate etching

Cyclic process:

- MoF_6/H_2O half-reactions saturate without spontaneous etching
- Converted surface to Mo-O before fluorination

Oxidation

Mechanism:

- O_2 activation depends on site geometry
- SO_x loss competes with oxygen retention

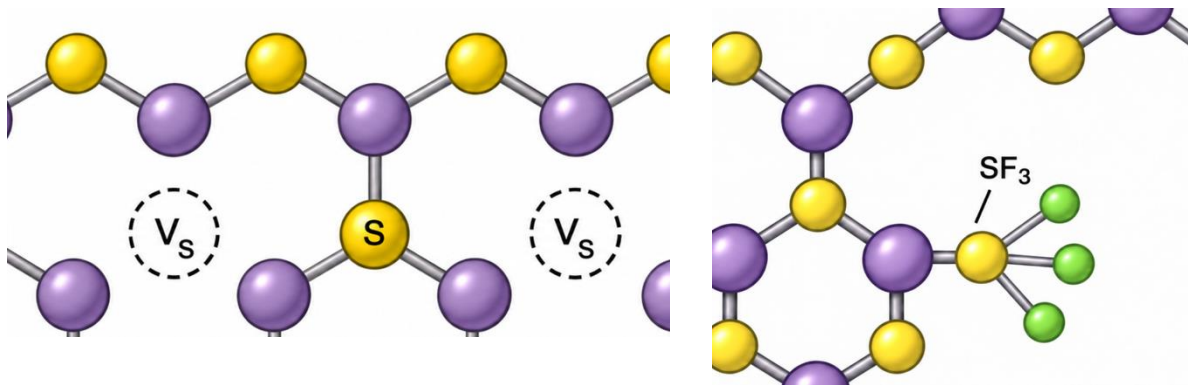
Cyclic process:

- $O\bullet/HCOOH$ remove ~ 1 layer per cycle
- Lower temperatures prevented spontaneous etching, allowing $HCOOH$ to do the removal

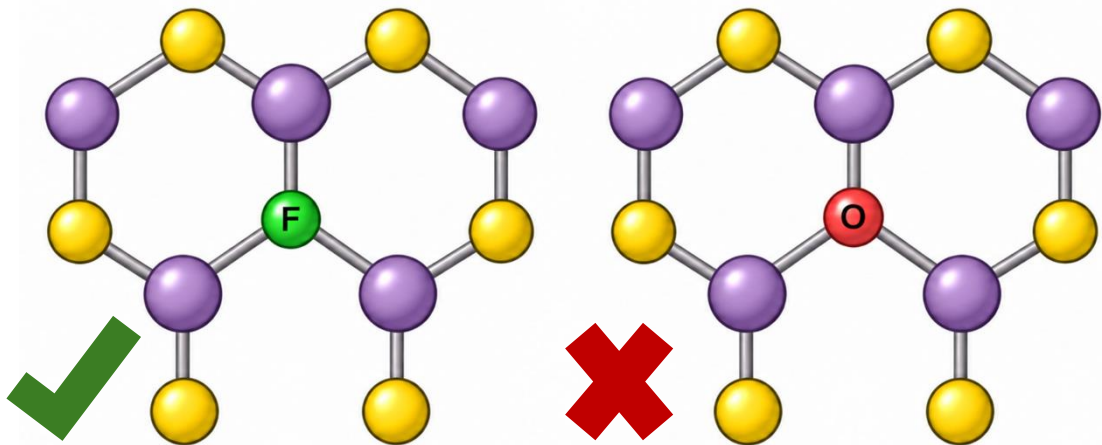
Different mechanisms create different control challenges

What controls etching?

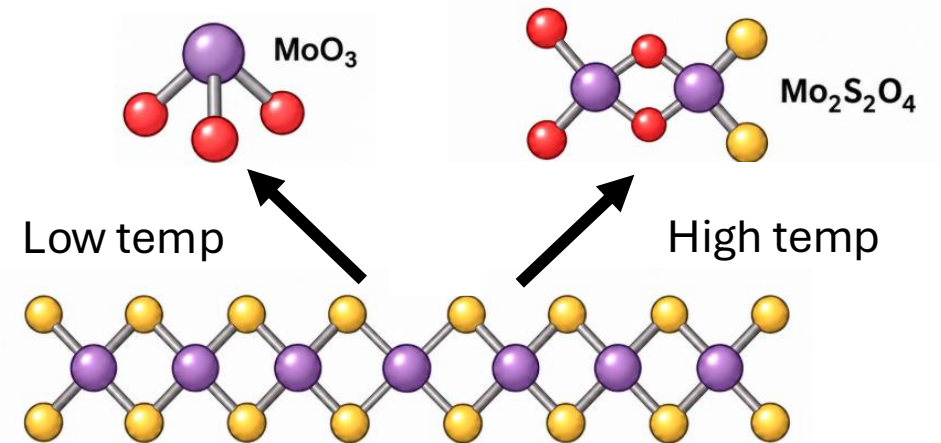
Site geometry



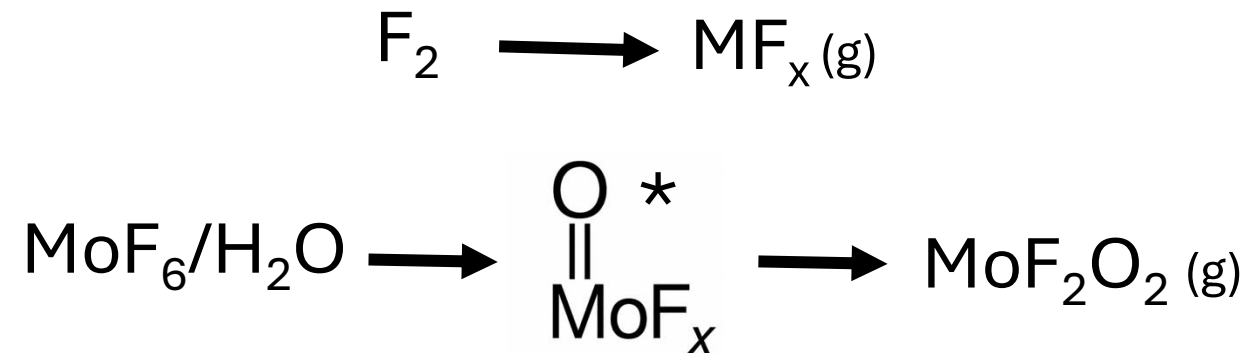
Surface state



Reaction conditions



Pre-cursor chemistry

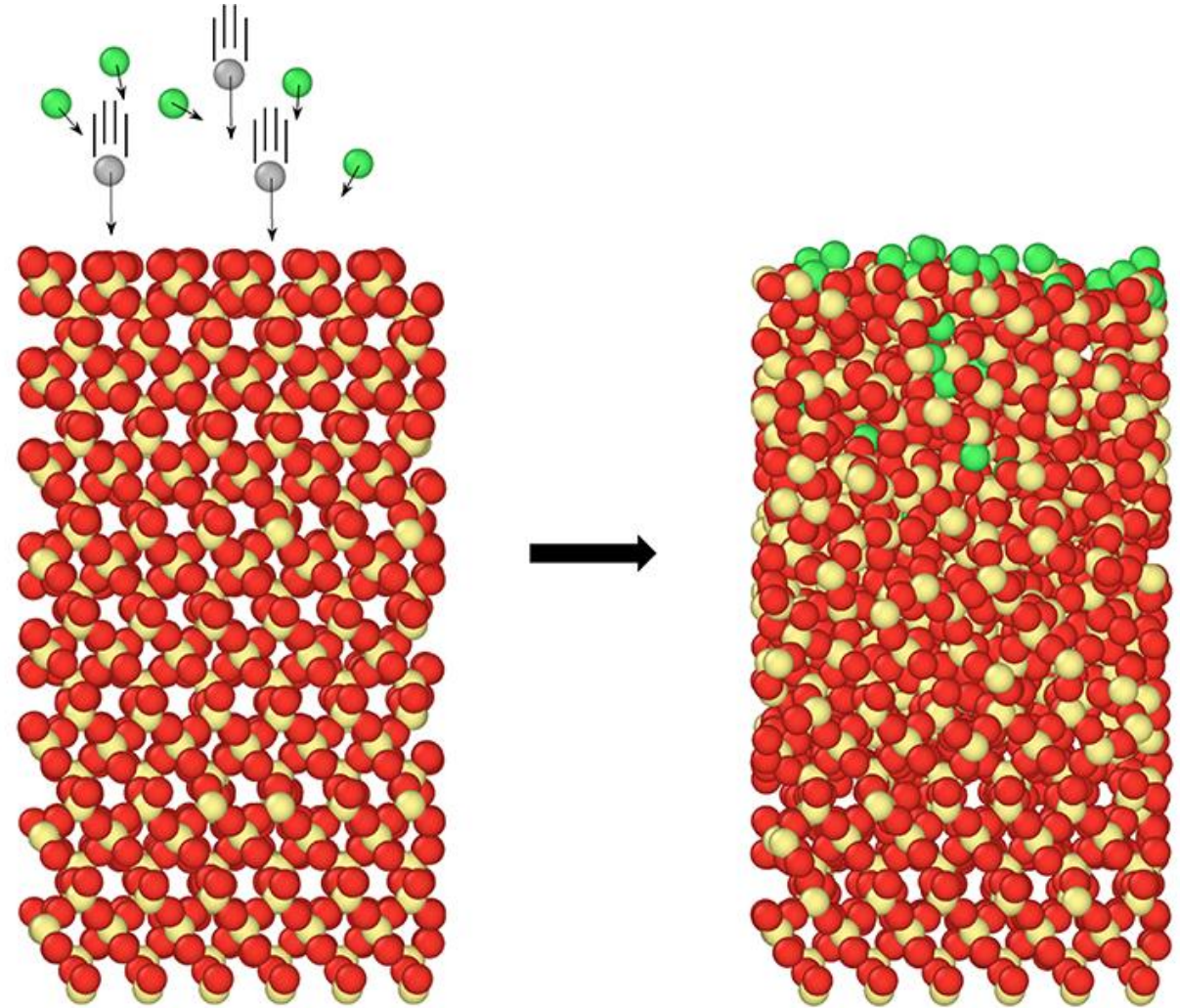


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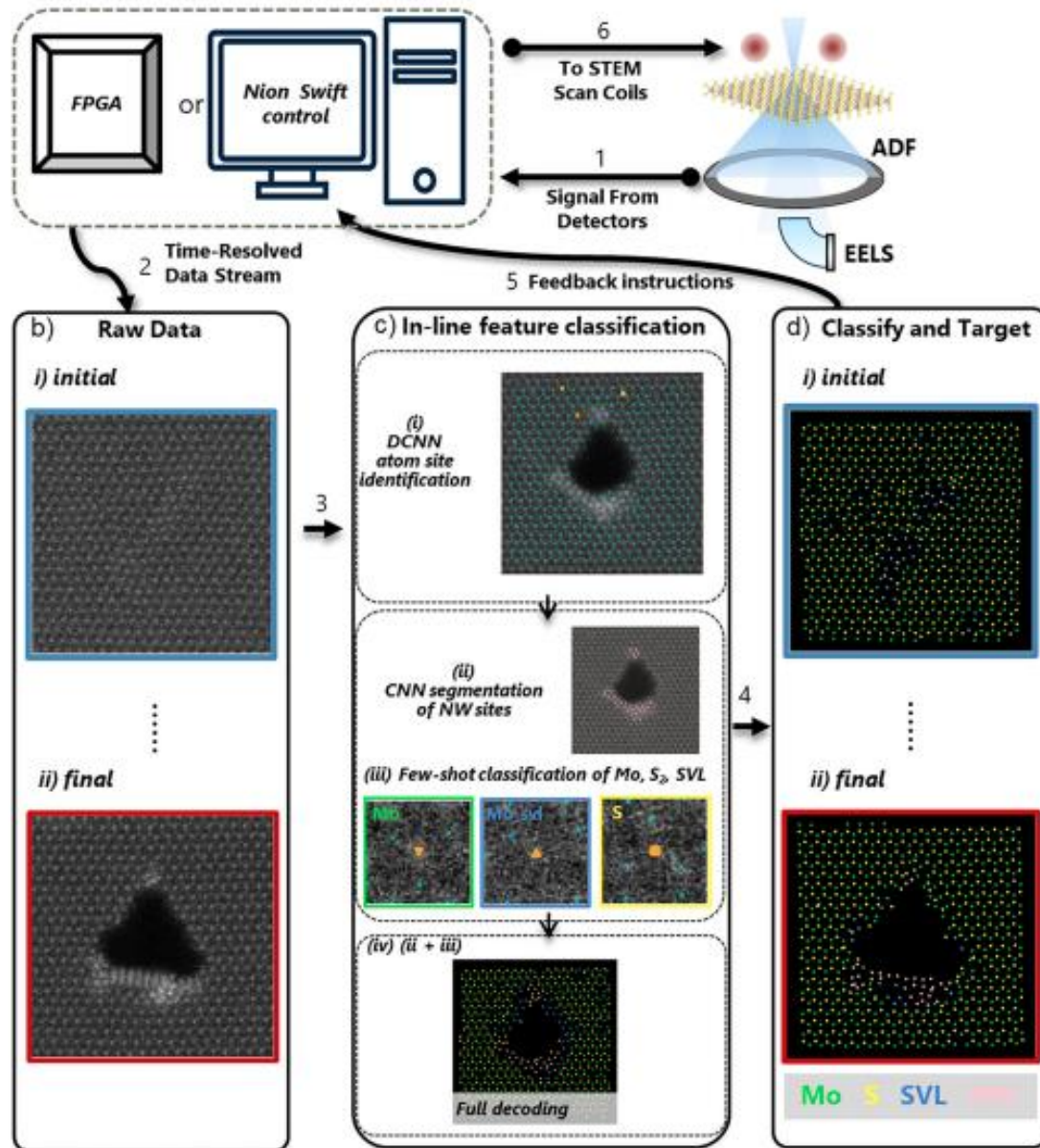
Simulating etching with MLIPs

- Machine-learned interatomic potentials (MLIPs)
- Near-DFT accuracy at larger scales
- Could simulate MoS_2 defects, repeated exposures, and etch propagation
- Active learning targets uncertain chemistry



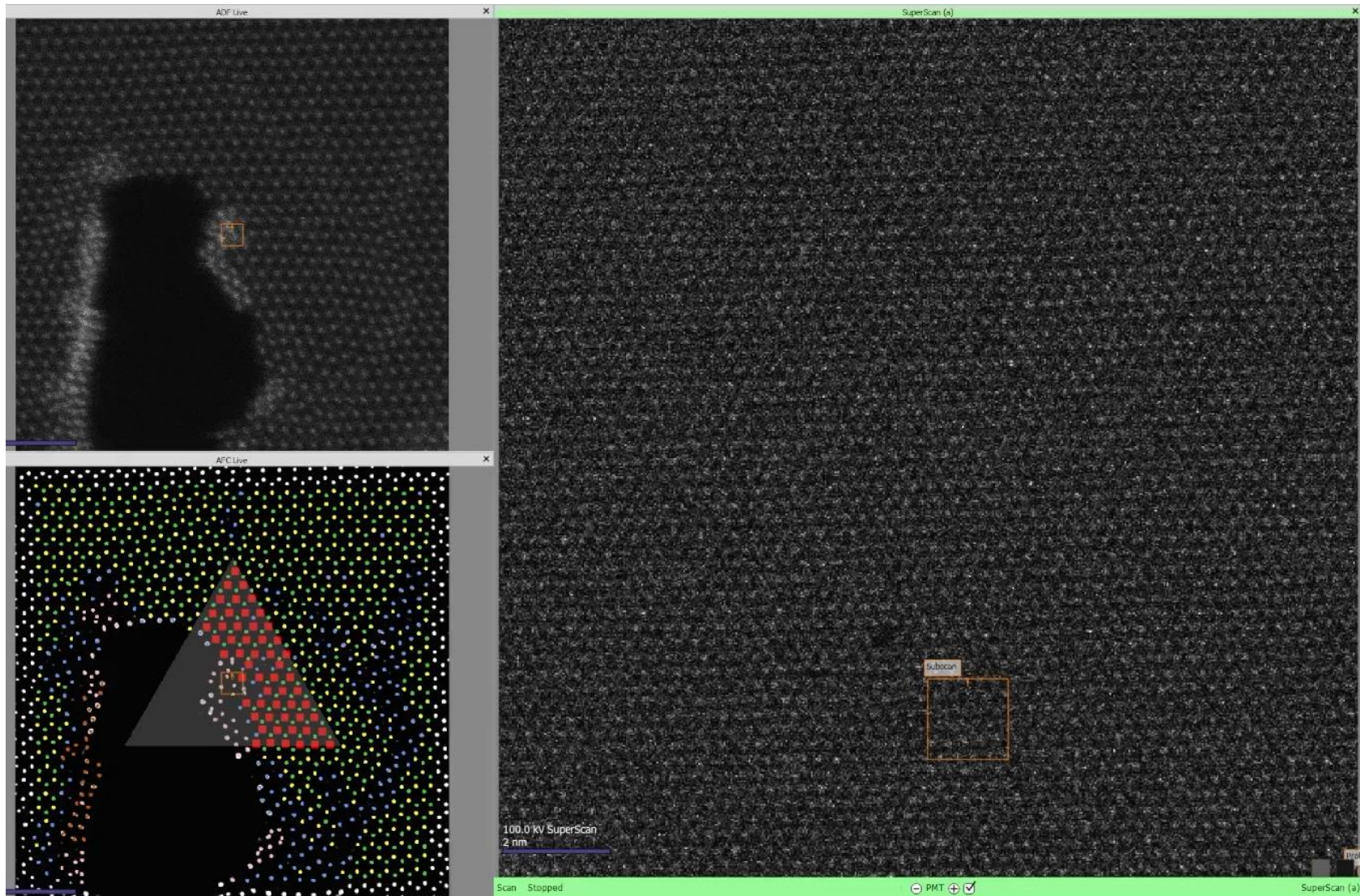
Example: MLIP simulation of ion-enhanced fluorine Si/SiO₂ etching

ML-guided electron-beam fabrication



- STEM beam selectively removes S atoms from monolayer MoS₂
- Trained ML model converts each image into a map of atoms, vacancies, and evolving edges
- Observed structure determines which sites the beam targets next
- Repeating the feedback loop guides fabrication toward a predefined structure

ML-guided electron-beam fabrication



Outlook

- Closer links between simulation and experiment
 - Test predicted intermediates and removal pathways against measurements
- More realistic simulations
 - Larger systems, longer timescales
- Feedback-guided processing
 - Use observed surface state to adjust processing conditions live with ML

References

1. Lan, H.-Y.; Yang, S.-H.; Cho, Y.; Tan, Y.; Cai, J.; Sun, Z.; Li, C.; Huang, L.-Y.; Wan, Y.; Li, L.-J.; Beechem, T.; Appenzeller, J.; Chen, Z. Scaling Two-Dimensional Semiconductor Nanoribbons for High-Performance Electronics. *Nat. Commun.* **2026**, *17*, 7386. DOI: 10.1038/s41467-026-74342-z.
2. Ossila. Molybdenum Disulfide (MoS₂): Theory & Applications. <https://www.ossila.com/pages/molybdenum-disulfide-mos2> (accessed 2026-09-01).
3. Kessels, E. Atomic Layer Etching Turns 32.5 Years Old!—A Good Occasion to Share an ALE Timeline and an Animated Version of the ALE Periodic Table. *AtomicLimits*, March 2, 2020. <https://www.atomiclimits.com/2020/03/02/atomic-layer-etching-turns-32-5-years-old-a-good-occasion-to-share-an-ale-timeline-and-an-animated-version-of-the-ale-periodic-table/> (accessed 2026-09-01).
4. Farigliano, L. M.; Paredes-Olivera, P. A.; Patrito, E. M. Ab-Initio Molecular Dynamics Simulations of the Reactivity of MoS₂ towards F₂ Molecules: Implications for Etching Processes. *Appl. Surf. Sci.* **2023**, *607*, 154637. DOI: 10.1016/j.apsusc.2022.154637.
5. Soares, J.; Mane, A. U.; Choudhury, D.; Letourneau, S.; Hues, S. M.; Elam, J. W.; Graugnard, E. Thermal Atomic Layer Etching of MoS₂ Using MoF₆ and H₂O. *Chem. Mater.* **2023**, *35*, 927–936. DOI: 10.1021/acs.chemmater.2c02549.
6. Gellerup, B. S.; Lasseeter, J. C.; Kang, K.; Xiao, K.; Zarkadoula, E.; Huang, J. J.; Retterer, S. T.; Randolph, S. J.; Rack, P. D. Nanoscale Editing of Multi- and Single-Layer Tungsten Disulfide via Gas-Assisted Focused Electron Beam Induced Etching for Device Prototyping. *Adv. Funct. Mater.* **2026**, *36*, e07816. DOI: 10.1002/adfm.202507816.
7. Farigliano, L. M.; Paredes-Olivera, P. A.; Patrito, E. M. Initial Steps of Oxidative Etching of MoS₂ Basal Plane Induced by O₂. *J. Phys. Chem. C* **2020**, *124*, 13177–13186. DOI: 10.1021/acs.jpcc.0c02141.
8. Farigliano, L. M.; Paredes-Olivera, P. A.; Patrito, E. M. Oxidative Etching of S-Vacancy Defective MoS₂ Monolayer upon Reaction with O₂. *Phys. Chem. Chem. Phys.* **2021**, *23*, 10225–10235. DOI: 10.1039/D0CP06502A.
9. Farigliano, L. M.; Monzón-Somazzi, C. M.; Navarro-Gamarra, K. E.; Paredes-Olivera, P. A.; Patrito, E. M. Oxidation of Mo-Terminated Pit Edges in MoS₂ Monolayer. *J. Phys. Chem. C* **2024**, *128*, 15852–15860. DOI: 10.1021/acs.jpcc.4c01781.
10. Sovizi, S.; Tosoni, S.; Szoszkiewicz, R. MoS₂ Oxidative Etching Caught in the Act: Formation of Single (MoO₃)_n Molecules. *Nanoscale Adv.* **2022**, *4*, 4517–4525. DOI: 10.1039/D2NA00374K.
11. Reidy, K.; Mortelmans, W.; Jo, S. S.; Penn, A. N.; Foucher, A. C.; Liu, Z.; Cai, T.; Wang, B.; Ross, F. M.; Jaramillo, R. Atomic-Scale Mechanisms of MoS₂ Oxidation for Kinetic Control of MoS₂/MoO₃ Interfaces. *Nano Lett.* **2023**, *23*, 5894–5901. DOI: 10.1021/acs.nanolett.3c00303.
12. Han, H.; Kang, J.; Kim, J.; Jeong, S.; Kim, S.; Oh, G.; Ko, D.; Son, S.; Choi, H.; Kim, D.; Lee, M.; Kim, T.; Kim, J.; Yeom, G. Isotropic Thermal Atomic Layer Etching of 2D MoS₂ Using Formic Acid Vapor. *Surf. Interfaces* **2026**, *84*, 108587. DOI: 10.1016/j.surfin.2026.108587.
13. Kounis-Melas, A.; Panagiotopoulos, A. Z.; Graves, D. B. Development of a Deep Potential Model for F and CF₂ Etching of Si and SiO₂. *J. Vac. Sci. Technol. A* **2026**, *44*, 043205. DOI: 10.1116/6.0005504.
14. Wu, Z.; Roccapiore, K. M.; Ghosh, A.; Xiao, K.; Unocic, R. R.; Jesse, S.; Vasudevan, R.; Boebinger, M. G. Autonomous Fabrication of Tailored Defect Structures in 2D Materials Using Machine Learning-Enabled Scanning Transmission Electron Microscopy. *npj Comput. Mater.* **2026**, *12*, 186. DOI: 10.1038/s41524-026-02025-w.



Thank you for your attention!

